

Empowering Women, Securing Food
How Patriarchal Norms and Marital Status Shape
Women's Access to Resources and Food Security
in Syria

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List of Abbreviations

FAO	Food Agriculture Organization
FCS	Food Consumption Score
FHHs	Female-Headed Households
HH	Household Head
HDDS	Household Dietary Diversity
ISDC	International Security Development Center
MHHs	Male-Headed Households
OLS	Ordinary Least Squares
RCSI	Reduced Coping Strategy Index
SDG	Sustainable Development Goals
USAID	United States Agency for International Development
WFP	World Food Programme

Executive Summary

This thesis examines the role of women in securing household food security in government-controlled areas of Syria and how this role is shaped by patriarchal norms, conflict and structural inequalities. In such fragile settings, achieving SDG2 (Zero Hunger) requires more than food aid, it demands a gender-sensitive understanding of how power, resources and responsibilities are distributed within households. Following the Arab uprising in 2011, many Syrian women assumed greater household duties due to the loss or displacement of male relatives. Yet this increase in responsibility has not been matched by greater empowerment. Women remain constrained by limited access to land, education and legal rights. While empowerment is widely seen as a pathway to food security, this thesis argues that in patriarchal, conflict-affected settings, it cannot be meaningfully understood without considering marital status, which shapes women's access to resources through gender norms and social structures.

Using original data from 932 female-headed households (FHHs) across five Syrian governorates, the study explores how land ownership and education affect food security, measured by the Food Consumption Score (FCS), Household Dietary Diversity Score (HDDS) and Reduced Coping Strategy Index (RCSI). The analysis tests both direct effects and interactions between marital status and these empowerment resources using nine pooled Ordinary Least Squares (OLS) models with clustered standard errors at the household level. Findings show that widowed women face higher food insecurity despite reporting greater decision-making authority, highlighting that autonomy alone is insufficient without access to resources. While married women benefit from male-mediated access to land and services, widows remain marginalized. However, when widows do gain access to resources, such as land, their household food security improves significantly, suggesting they benefit more than married women constrained by gendered household dynamics. Education, by contrast, offers limited protection due to systemic barriers like restricted mobility and limited employment opportunities.

Beyond its empirical contribution, this study reframes female headship not as a proxy for empowerment but as a condition that often masks deeper vulnerability. It highlights the need to understand empowerment as relational and resource-dependent and to recognize resilience as a key lens for analyzing how life shocks, like widowhood, reshape women's capacity to provide for their households.

1. Introduction

The 2030 Agenda for Sustainable Development identifies the eradication of hunger as a global priority. SDG2 calls on governments and international actors to end hunger, achieve food security and promote sustainable agriculture (United Nations, n.d.). Yet, despite extensive commitments, hunger remains a critical and unresolved global crisis. In 2020 alone, more than 811 million people were food insecure – a sharp increase driven by conflict, economic instability and climate change (FAO et al., 2021).

This crisis is especially acute in conflict-affected contexts such as Syria, where more than a decade of war has devastated livelihoods, fractured state institutions and intensified structural inequalities (Doocy et al., 2017). In such settings, where food systems are highly fragile and access is shaped by overlapping political and social disruptions, household food security cannot be addressed through food aid alone. It requires an understanding of the social dynamics that influence how food is accessed, controlled and distributed within households. Among these dynamics, gender plays a central role. Empowering women has been widely recognized as a strategic pathway to improving food security, particularly in low-income, agrarian and crisis-affected settings (Clement et al., 2019; Essilfie et al., 2024). Even being at the forefront of food production and provisioning, in patriarchal societies women are often expected to fulfill these roles without secure access to land, income, education or legal protections (Laar and Aryeetey, 2014). In fragile contexts, the link between gender equality and food security is especially stark. For instance, the Syrian uprising in 2011 has displaced or killed thousands of men, compelling women to step into new roles as primary caregivers, income earners and household heads (HHs) without corresponding legal reforms or structural support (UNDP, 2018; SCPR, 2013). This led Syrian women often to carry full household responsibility without the rights traditionally granted to male HHs highlighting a key misconception in both policy and research: holding household responsibility does not inherently equate to empowerment. As Kabeer (1999) argues, empowerment is a multidimensional process involving access to resources, agency and the ability to achieve desired outcomes – such as improved household food security. These elements are deeply interconnected. In patriarchal societies like Syria, where the distribution of resources is

mediated by male authority figures (fathers, husbands, sons) agency alone is insufficient. A woman may take part in decision-making, but without a real control over land, income or education, she cannot act meaningfully on them to improve household welfare (Quisumbing et al., 1996).

Building on this, it is essential to consider how marital status influences the relationship between empowerment and household food security. Despite the growing literature on this topic, marital status remains an under-theorized dimension. Research often treats female heads as a uniform group, overlooking how empowerment differs between married and widowed women. Indeed, marital status shapes household roles, legal recognition and social legitimacy (Evanson, 2016; IICISAR, 2023).

This thesis responds to this gap by investigating how marital status mediates the relationship between women's empowerment and household food security in conflict-affected Syria. Rather than treating marital status as a simple demographic variable, this research reframes it as a structural condition that shapes access to resources, agency and achievements. The study asks whether widowed women, expected to be more autonomous, are in fact more empowered or simply more burdened.

Using original panel data collected from five Syrian governorates between 2023 and 2024, this study explores how two key empowerment-related resources such as land ownership and education, affect household food security across married and widowed FHHs. Food security is measured through three internationally recognized indicators: the FCS, HDDS and RCSI (WFP, 2008; Maxwell & Caldwell, 2008).

To guide this inquiry, the thesis addresses the following research questions:

1. How does marital status shape women's empowerment and household food security among FHHs in Syria?
2. To what extent do empowerment-related resources like land ownership and education influence household food security outcomes in patriarchal settings?
3. Do widowed women benefit more from empowerment-related resources than married women, and if so, how does this reflect structural inequalities in resource access?

To answer these questions, the analysis draws on primary household-level panel data collected across five Syrian governorates and employs nine pooled OLS regression models with clustered standard errors – three for each food security outcome. The models are structured to test direct effects (Model I), as well as interactions between marital status and land ownership (Model II) and marital status and education (Model III), corresponding to the study’s five hypotheses.

The findings reveal that while widowed women report greater decision-making authority (Figure 8), they also experience higher food insecurity due to reduced access to resources. Land ownership emerges as a compensatory factor, particularly in improving HDDS among widowed women. By contrast, education does not appear to provide the same protective effects, likely due to persistent structural barriers such as unemployment, mobility restrictions and gendered exclusion from economic and civic life.

In conclusion, this research argues that in crisis settings like Syria, empowerment must be understood as relational, resource-based and contextually constrained. Marital status remains a central axis through which gendered power is exercised and limited. As such, achieving SDG2 is inseparable from realizing SDG5 (Gender Equality) – but only if empowerment is matched with access to the resources and rights that make food security possible.

2. Theoretical Framework

2.1 Food Security

Food security is a pressing global challenge. In 2020, over 811 million people were estimated to be food insecure (FAO et al., 2021). This concern is even more acute in fragile settings like Syria, where social, political and environmental shocks intersect to exacerbate vulnerability (World Bank, 2022).

According to FAO (1996), “Food security exists when all people, at all times, have physical and economic access to sufficient, safe and nutritious food to meet their dietary

needs and food preferences for an active and healthy life.” Building on this definition, food security is understood through four core dimensions (Essilfie et al., 2024, p. 3):

- Availability: The physical presence of food in a region or household.
- Access: The ability of individuals to obtain food through economic and social means.
- Utilization: The nutritional value and safe consumption of food, ensuring dietary adequacy.
- Stability: The sustained access to adequate food over time, especially in the face of shocks.

Especially in conflict-affected settings, it is essential to analyze all dimensions of food security comprehensively. While access to a sufficient and nutritious diet is crucial, the dimension of stability becomes particularly vital, as it determines whether households can maintain consistent access to food amid disruption, displacement or economic collapse. In the Syrian context – where instability is protracted and systemic – long-term food security hinges not only on availability and access, but on the household’s capacity to sustain these conditions over time.

2.1.1 Gendered Dynamics of Household Food Security

Understanding food security at the household level requires attention to gendered disparities in resource access and control (Doss, 2013; Duflo, 2012). In patriarchal societies like Syria, men and women experience fundamentally different realities in terms of income earning, asset ownership, food consumption and decision-making authority. Women frequently bear the responsibility for ensuring household food supply, even while being denied access to productive resources (Jemaneh & Shibeshi, 2023, p. 2). Syrian women’s experiences reflect a global pattern: women contribute to 45-80% of household food production – rising to 70-80% in sub-Saharan Africa, 65% in Asia, and 45% in Latin America and the Caribbean – while consistently facing structural barriers to control over agricultural inputs and outputs (Quisumbing et al., 1996, p. 1). This contradiction of high responsibility with low control reveals that food insecurity is fundamentally a matter of power and availability. Research demonstrates that women’s vulnerability stems not from

any lack of capacity, but rather from structural disadvantages embedded within legal and social systems that systematically limit their authority and decision-making power (Clement et al., 2019). In Syria specifically, these dynamics are exacerbated by legal frameworks and cultural norms that entrench male guardianship over women's property, inheritance and financial autonomy (UNDP, 2018). The consequences are stark: FHHs face disproportionate hardship, being twice as likely as male-headed households to report a complete inability to meet basic needs (IICISAR, 2023).

Despite these constraints, women often demonstrate a stronger commitment to household nutrition. Social expectations around caregiving, coupled with the nature of women's income (typically smaller and more frequent), result in women allocating a greater share of their resources to food, health and child welfare (Quisumbing & Maluccio, 2003). They also play a key role in ensuring food quality and diversity, engaging not only with caloric sufficiency but also with nutritional adequacy (Colecraft et al., 2006). In this regard, education functions as a critical enabler: women with higher educational attainment are better equipped to make informed choices that enhance household food security (Adepoju et al. 2015).

Taken together, these insights underscore that food security is not only gendered but deeply political. While women's contributions to household food provisioning are substantial, their ability to act effectively is systematically constrained by structural inequalities (UNDP, 2018). This is particularly evident in contexts like Syria, where even becoming the HH does not necessarily equate to genuine empowerment. FHHs often face harsher barriers and greater vulnerabilities than their male-headed counterparts, revealing that holding formal decision-making authority is not sufficient when access to resources remains mediated by social and legal structures (IICISAR, 2023). As this research will explore, widowed women frequently lack the social legitimacy and resource access needed to translate complete control over household decisions into improved food security outcomes. Thus, food security must be understood not simply as a matter of access or availability, but as fundamentally tied to processes of empowerment and disempowerment – processes that are relational, fragile and deeply embedded within patriarchal norms.

2.2 Empowerment as a Pathway to Food Security

In this context, women's empowerment emerges as a key pathway to addressing household food insecurity. Empowerment enables women to exercise greater control over resources, make strategic decisions and mitigate the effects of shocks. Empirical studies confirm that more empowered women are more likely to ensure household food security and child well-being (Essilfie et al., 2024). While the specific drivers of empowerment vary across contexts – ranging from education to land rights to social norms – the outcomes are broadly consistent: empowered women feed their families better (Clement et al., 2019).

Recent literature (Lufuke et al., 2022; Cornwall, 2016) identifies three dimensions of empowerment that are particularly relevant to food security:

- Physical and Economic Agency – Control over income, land and household resources enhances food access and prioritization.
- Socio-cultural Factors – Education and mobility enable women to make informed nutritional and financial decisions.
- Cultural Values – Societal recognition of women's authority strengthens their role in resource allocation.

Following this vision, in conflict-affected areas like Syria – where food insecurity and recurring emergencies are endemic – supporting women's empowerment is not only essential for gender equity, but also a strategic pathway to strengthening household and community resilience. Women's adaptive roles in ensuring household nutrition, managing scarce resources and navigating instability make them vital agents in fragile settings (Essilfie et al., 2024; Clement et al., 2019). However, the meaning and effectiveness of empowerment are profoundly shaped by prevailing gender norms and social hierarchies. As a matter of fact, a woman's perceived legitimacy and access to resources are often mediated by her societal and marital status.

2.3 Women's Empowerment and Marital Status

Women's empowerment is widely recognized as a dynamic, multi-dimensional and deeply context-dependent process (Le & Nguyen, 2020, p.1). It is not a fixed or linear state, but rather a progressive shift from disempowerment to empowerment, shaped by individual agency, access to resources and the capacity to achieve valued outcomes (Kabeer, 1999). At the heart of this transformation lies a woman's ability to make strategic life choices.

Kabeer's (1999) influential framework breaks empowerment down into three interrelated dimensions:

- Resources – the material, human and social assets that form the preconditions necessary to exercise choice.
- Agency – the ability to define one's goals and act upon them.
- Achievements – the outcomes that reflect whether choices have improved an individual's or a household's well-being.

These dimensions present empowerment as a process where resources enable agency, agency fuels decision-making and achievements reinforce further empowerment. At the core of this process lies the capacity to make choices that are not only available but also real and meaningful. Critically, for choices to be empowering, women must recognize that alternatives exist and believe they are entitled to pursue them. Here, Freire's (1996) concept of *critical consciousness* becomes essential: empowerment begins when women identify the structural barriers – legal, political and cultural – that restrict them and develop an awareness of their rights and capacity to resist and redefine these limitations (Leder, 2016). In deeply patriarchal societies, this basic process – what Rowlands (1998) calls power within – is shaped by women's internalized norms and social positioning, and marital status plays a crucial role. Married women often derive social legitimacy from their association with a male partner, which can obscure or even suppress their awareness of structural inequalities by reinforcing dependence and limiting the perceived need to challenge the status quo (Leder, 2016). In contrast, widowed women – while more socially marginalized – may be more likely to confront structural constraints and claim decision-making roles out of necessity (Carr, 2004). However, this internal transformation alone is not sufficient.

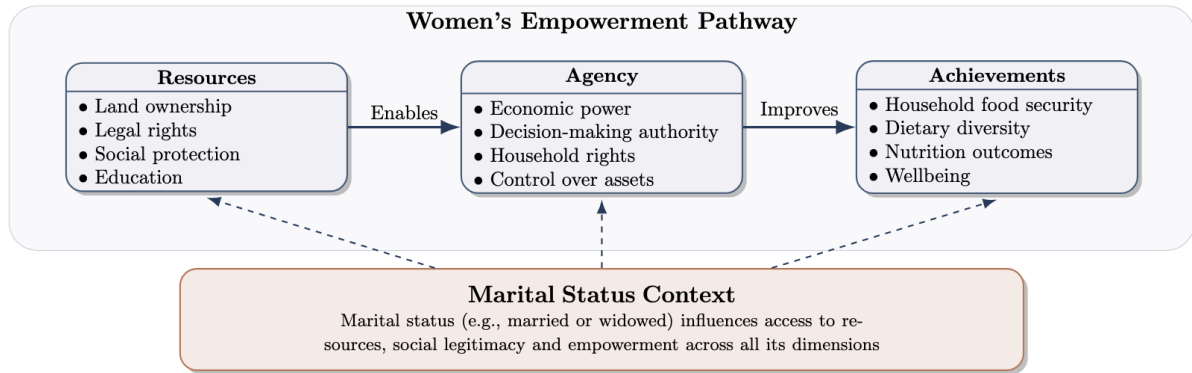
Empowerment also requires relational and structural dimensions. Rowlands' (1998) expanded typology of power outlines two other forms of power:

- Power over – the institutional and cultural constraints that limit women's access to resources and decision-making spaces.
- Power with – the importance of solidarity and collective identity, especially for women whose social legitimacy may be questioned based on their marital status.

These dynamics are especially relevant in Syria, where women's access to rights, assets and opportunities is often mediated through male relatives (Evanson, 2016), and marital status plays a critical role in reshaping the gendered distribution of household power and resources (Dercon & Krishnan, 2000; Van de Walle, 2011). According to the Under Syria's Personal Status Law, a wife's legal status is subordinate to her husbands in matters of marriage, divorce, custody and inheritance, requiring male guardianship for various legal transactions (Bellafronto, 2005). Due to this structure, widows often face compounded disadvantages: despite serving as primary caregivers and households, they can face stigma, exclusion from inheritance and formal safety nets and marginalization from social and institutional networks (Dercon & Krishnan, 2000; Van de Walle, 2011). In contrast, married women tend to maintain stronger access to community networks and legal protections through their association with a male partner, benefiting from social recognition and institutional support. Therefore, the sudden absence of a male partner often strips women not only of material support but also of their social legitimacy, reinforcing systemic disempowerment and exposing them to deeper emotional and legal hardships, particularly in settings where laws around inheritance, guardianship and mobility continue to favor male authority.

For this reason, in patrilineal societies, being a female HH does not automatically imply genuine empowerment. Marital status must be seen not as a background demographic trait, but as a key determinant of women's access to resources and social legitimacy when assessing empowerment and household food security. The Syrian conflict has further disrupted traditional male roles forcing many women to take on provider and decision-making responsibilities without any legal and structural reform. (Alsaba & Kapilashrami, 2016, p. 9). Building on this social shift, this thesis reframes marital status as a core structural factor that both reflects and reshapes empowerment trajectories among FHHs, consequently influencing food security.

Figure 1. Women's Empowerment and Food Security pathways mediated by Marital Status



3. Prior Research

3.1 Women's Empowerment and Household Food Security: Global Evidence

Over the past two decades, several scholars have underscored the critical relationship between women's empowerment and household food security. This connection is particularly present in low- and middle-income countries, where women play a central role in both agricultural production and household provisioning (Duflo, 2012). Yet, despite their centrality in food systems, women's ability to ensure adequate food for their families is often constrained by entrenched structural inequalities. These include limited access to education, land and credit, as well as restrictive gender norms that govern household decision-making and resource distribution. Laar and Aryeetey (2014) attributed women's food insecurity to deeply rooted socio-economic hierarchies and cultural expectations that position women as caregivers with limited household decision-making power. In many settings, these inequalities often manifest at the intra-household level. For example, Colecraft et al. (2006) found that in some Ghanaian households, men are routinely prioritized in the allocation of nutrient-rich foods, while women often skip meals to ensure their children are fed.

Despite these structural barriers, empirical research consistently shows that increased access to education, income and household authority enables women to significantly improve food security outcomes – particularly with respect to dietary diversity and child nutrition. Essilfie et al. (2024) found that in Northern Ghana, women’s education and household decision-making power were strongly associated with reduced food insecurity. Quisumbing and Maluccio (2003) similarly demonstrated that income earned by women is more likely to be spent on food, education and healthcare, contributing to the overall household welfare, while Olumakaiye and Ajayi (2006) demonstrated that women’s education in Nigeria correlates with higher dietary diversity at the household level. These studies collectively affirm that empowering women by enhancing their control over resources and participation in household decisions can yield substantial improvements in food security at the household level. However, empowerment is a multidimensional process and cannot be reduced to a single domain, such as economic participation or resource control (Jemaneh & Shibeshi, 2023). As Malapit and Quisumbing (2015) argue, although women’s participation in financial decision-making is linked to improved dietary diversity, it does not always translate into better nutritional outcomes. This is often due to mediating factors such as poor sanitation or limited access to healthcare – conditions that fall outside the direct scope of women’s control but nonetheless exert a significant influence on women’s access to resources and decision-making power. These findings underscored the need to analyze empowerment as a multifaceted and context-sensitive process, rather than reducing it to economic or decision-making indicators alone.

For this reason, several scholars have based their concept of empowerment on Kabeer’s notion of a process, emphasizing that *access* to resources does not automatically translate into meaningful *agency*. Smith et al. (2003), for example, found that while women with higher social status may manage household resources more effectively and improve child nutrition, their influence often remains limited to maternal and child outcomes, without challenging or reshaping broader intra-household power dynamics. Similarly, Rehman et al. (2019) linked women’s land rights in Pakistan to improved household welfare and nutritional outcomes, but their study also highlighted that legal ownership alone does not guarantee empowerment unless it is accompanied by the ability to exercise control and decision-making over those assets.

Recognizing the multifaceted nature of empowerment – where women’s autonomy must be supported by broader structural and cultural transformations to be effective – recent studies have conceptualized empowerment as a dynamic and relational process shaped by legal frameworks, social norms and intra-household power dynamics. For instance, Wei et al. (2021) offer a particularly comprehensive critique of earlier literature that tends to frame it predominantly in economic terms. By conceptualizing empowerment across six key dimensions – legal rights, leadership, infrastructure, information and communication technologies (ICTs), social norms and access to knowledge – they situate the concept within a broader structural and relational context. Their study emphasizes that even when rural Bangladeshi women participate in food-related decision-making, deeply internalized stigma and low self-esteem can significantly undermine their capacity to act as autonomous agents. These findings challenge reductive definitions of empowerment based solely on resource ownership or formal authority and instead call for an expanded understanding that incorporates relational, psychological, institutional and cultural dimensions, such as gendered intra-household food dynamics.

In response to these concerns, the International Food Policy Research Institute developed the Women’s Empowerment in Agriculture Index (WEAI) to capture empowerment more holistically across five domains: production, access to resources, income control, leadership and time use (CGIAR, n.d.). Sraboni et al. (2014) applied the WEAI in Bangladesh, confirming a positive correlation between greater empowerment and improved HDDS. However, their findings also highlighted critical limitations: despite its multidimensional approach, the WEAI struggled to capture subjective, relational and anticipatory aspects of empowerment – particularly how life shocks, such as partner loss, reshape women’s social and economic trajectories.

While indexes like the WEAI provide valuable insights, they often conceptualize empowerment as a stable condition achieved through resource access or decision-making authority, overlooking its fluid and context-dependent nature. Crucially, marital status continues to be treated merely as a background characteristic rather than as a structural determinant that profoundly shapes women’s agency, access to resources and resilience. In fragile contexts like Syria, where women’s legitimacy remains closely tied to male association, empowerment must be understood relationally. Widowhood may appear to grant autonomy through household headship but often limits access to resources, rights

and networks. Married women, meanwhile, may access institutions more easily but face restricted agency. Yet, despite these dynamics, marital status remains largely overlooked in empowerment and food security research – revealing a critical gap this thesis seeks to address.

3.2 Food Security in Conflict and Protracted Crisis Contexts

While the literature establishes women's empowerment as crucial for household food security in fragile contexts (Doss, 2013; Duflo, 2012), understanding these dynamics in conflict-affected settings requires additional consideration of how violence, displacement and institutional breakdown reshape gender roles and power relations. In these contexts, women's vulnerability often intensifies even as their responsibilities expand (SCPR, 2013, p. 15) – a paradox with profound implications for household resilience and food security.

In conflict-affected settings, food insecurity cannot be analyzed solely through the lenses of poverty or gender inequality. Much of the existing literature in this field focuses on the household welfare, with a strong emphasis on humanitarian outcomes such as resource availability, infrastructure damage or access to basic services. For example, research on Gaza assessed post-conflict household resilience largely through indicators like physical destruction and service delivery (Brück et al., 2019). While valuable, this humanitarian framing often overlooks the sociological dimensions of resilience – particularly the gendered transformations that take place within households and communities during and after conflict. It fails to account for how particularly vulnerable groups – such as women, who have been consistently shown to play a central role in mitigating food insecurity (Essilfie et al., 2024) – are themselves impacted by shocks. In these contexts, the absence of male household members often compels women to assume new forms of agency – navigating increased responsibilities and decision-making roles within precarious and deeply unequal conditions (SCPR, 2013, p. 15).

Some scholars have begun to address this gap by examining the role of women in improving household food security within conflict-affected contexts. These contributions moved beyond macro-level analyses considering gendered dynamics within households.

However, the academic literature on this intersection remains notably limited – particularly in the case of Syria. For example, George et al. (2020) illustrate how, during prolonged conflict, women often assume new responsibilities in the absence of male household members, leading to reconfigurations of domestic authority and social expectations. Similarly, Agidew and Singh (2018) found that FHHs in conflict-affected areas of Ethiopia tended to achieve greater dietary diversity, suggesting a stronger focus on food quality. However, these same studies also reveal that despite women’s growing roles in household decision-making, their agency remains constrained by entrenched structural and cultural patriarchal norms (Agidew and Singh, 2018; SCPR, 2013, p. 18-19). As a result, these households often continue to experience higher levels of food insecurity due to persistent barriers in accessing financial and institutional support.

Critically, existing research on conflict zones focuses predominantly on humanitarian assistance delivery rather than exploring how women’s empowerment functions as a pathway toward sustainable food security. This represents a significant gap in understanding how to build local resilience beyond immediate aid provision. The limited literature rarely examines how marital status fundamentally shapes women’s power, agency and ability to ensure household food security.

Focusing on Syria, the literature remains extremely limited – likely due to restricted data access and ongoing insecurity. One of the few empirical studies on this topic is the one conducted by Doocy et al. (2017), who found that FHHs in government-controlled areas experienced higher levels of food insecurity than their male-headed counterparts, despite receiving more humanitarian assistance. These women relied more heavily on coping strategies, often sacrificing their own nutrition for the household’s welfare. However, the study did not investigate women’s empowerment as a potential pathway to improve household food security, nor did it explore how their marital status reshapes women’s agency, with implications for household resilience and long-term food access.

3.3 Research Contribution to the Literature

This thesis addresses critical gaps in the existing literature on women's empowerment and household food security in conflict-affected contexts, particularly in Syria.

First, this research advances theoretical understandings by examining how marital status mediates the relationship between women's empowerment and household food security in conflict settings. Rather than treating marital status as a mere demographic variable, this study reframes it as a structural determinant that fundamentally shapes women's pathways to empowerment and their ability to ensure household food security, something inadequately explored in settings where conflict intersects with entrenched patriarchal structures.

Second this research makes a unique contribution by utilizing primary data collected directly from Syria – an exceptional achievement given the severe challenges of conducting empirical research in conflict settings. This dataset enables grounded insights into Syrian household food security, moving beyond theoretical assumptions or reliance on proxy environments. Furthermore, the study stresses the need for interventions that consider cultural and social norms – particularly the roles of women in patriarchal societies – to ensure that resilience strategies are both contextually appropriate and sustainable.

Third, unlike previous studies that reduce food security to isolated metrics such as child nutrition or caloric intake, this research provides a comprehensive analysis across the four core dimensions of food security (availability, access, utilization and stability) specifically adapted to conflict-affected settings. This holistic framework captures the complex interplay between empowerment processes, structural inequalities and crisis dynamics in shaping household food security outcomes.

4. Gender, Power and Transformation in Syria's Historical Context

4.1 Women in Civil Society and Politics: A Historical Perspective

4.1.1 The Role of Women Before the Arab Spring (1946–2011)

To understand the impact of women's empowerment on household food security in Syria, it is essential to first situate Syrian women within their historical and cultural context. Women's roles have been shaped by overlapping forces: legal frameworks, state ideologies, patriarchal norms and moments of national transformation. These forces have historically regulated not only women's rights and access to resources, but also their visibility and voice within public and private domains. The multidimensional nature of empowerment cannot be fully grasped without recognizing how these domains have been constructed, contested and redefined over time in Syria.

Following independence from French rule in 1946 (Encyclopædia Britannica, 2025), women gained the right to vote in 1949 through their participation in nationalist movements (Alsaba & Kapilashrami, 2016, p. 8). Yet, as Makdisi notes, women's contributions to national liberation came at a cost: to maintain alignment with broader state-building priorities, women were often compelled to suppress demands for gender-specific rights and autonomy (SCPR, 2023, p. 8; Sparre, 2008).

A critical turning point came with the rise of the Ba'ath Party in 1963, following a military coup that introduced sweeping social reforms. Universal policies promoted subsidized food prices, expanded education, improved healthcare and industrial development (SCPR, 2013, p. 29). Women particularly benefited from enhanced access to education and the labor market, with significant improvements in literacy rates and school enrollment for girls (Alsaba & Kapilashrami, 2016, p. 7). This era also witnessed the feminization of specific sectors, particularly education, where women increasingly assumed teaching and administrative roles (SCPR, 2013, p. 18). Nonetheless, while appearing progressive, the Ba'athist approach to gender equality was deeply instrumental. It was framed as a nationalist duty rather than a right (SCPR, 2013, p. 18). Rural women continued to face significant inequalities (Alsaba & Kapilashrami, 2016, p. 7), while

feminist activism was restricted through regime-controlled bodies such as the General Women's Union, reinforcing neo-patriarchal structures that subordinated women's roles to state agendas (Alsaba & Kapilashrami, 2016, p. 8-9).

In the late 1990s and early 2000s, following the end of the Cold War and the loss of key trading partners (Joya, 2012, p. 30), the Syrian government under Bashar al-Assad shifted toward neoliberal economic policies, disproportionately affecting women. Public support structures eroded, leading to declining healthcare access, increased school dropout rates among girls and stagnating political representation (Alsaba & Kapilashrami, 2016, p. 8). In 2003, Syria ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), appearing as a signal of commitment to improve women conditions. However, the government placed reservations on several key articles related to divorce, nationality, inheritance, reproductive rights and protections against child marriage; thereby perpetuating the enduring influence of patriarchal structures and the lack of political will to fully recognize women's right (Alsaba & Kapilashrami, 2016, p. 7). During this period, political and economic power remained concentrated among a coalition of the old state bourgeoisie, religious authorities, emergent corporate actors and militarized institutions (Alsaba & Kapilashrami, 2016, p. 8). Religious conservatism increasingly shaped public discourse, further reinforcing traditional gender roles and limiting women's participation (Joya, 2012, p. 31). Consequently, women's dependence on male family members deepened, diminishing their agency within both households and society. This reliance heightened their vulnerability during crises, particularly under economic shocks and political instability (Joya, 2012, p. 30). Despite these challenges, some indicators of educational investment remained strong: between 2005 and 2011, the literacy rate for individuals aged 15 and above was about 78%, and in 2009, Syria invested 5.1% of its GDP in public education sector. However, these investments failed to achieve equitable access or gender parity, especially for rural and low-income communities, where systemic barriers persisted (Zeno, 2021, p. 285).

4.1.2 The Role of Women During the Arab Spring (2011–2018)

The outbreak of the Syrian uprising in 2011 marked a critical juncture for women's political and social visibility. As anti-government protests spread across the country, Syrian women emerged as key figures in organizing, documenting and supporting the early non-violent resistance. They played significant roles in humanitarian aid, political activism, media work, and legal support, often assuming leadership positions in grassroots networks (SCPR, 2013, p. 15). This moment of civic engagement was viewed by many Syrian women as an opportunity to push forward long-suppressed socio-economic demands, which had been consistently sidelined under decades of Ba'athist rule (Hijazi, 2022).

In the early phase of the revolution, women were essential actors in the resistance, for their effectiveness and their ability to move across military checkpoints with less suspicion than men. For this reason, many women led local efforts in health, food and civil services, especially as state infrastructure collapsed (Alsaba & Kapilashrami, 2016, p. 9). Scholar Larbi Sadiki describes this phase as a time when women gained “acknowledgment and respect” from male peers, symbolically assuming the status of full citizens (Sadiki, 2014). However, these gains were short-lived. As the revolution militarized, women were marginalized again within political and military hierarchies and disproportionately affected by what was termed “cycled violence”. Alongside displacement, poverty and food insecurity, women faced escalating sexual violence, harassment, abductions and coerced marriages (SCPR, 2013, p. 15). Extremist ideologies and the lack of institutional protection exacerbated this reality. In regime-held areas, veiled women were harassed, while in opposition or ISIS-held regions, strict dress codes

and gender segregation were enforced. Women's bodies became politicized symbols, subject to humiliation or control, depending on the governing faction (SCPR, 2013, p. 17).

With the mass displacement, imprisonment or death of men, many women became sole HHs, assuming new socio-economic roles. However, this shift did not translate into structural empowerment. Women continued to face legal marginalization in property ownership, inheritance and guardianship over their children. In the absence of systemic change, these new responsibilities often intensified women's burden without increasing their agency or rights. Even those who joined armed factions did so under limited inclusion. Their involvement rarely challenged entrenched gender hierarchies and exposed them to further exploitation and violence (SCPR, 2013, p. 18-19).

4.2 The Role of Women in Contemporary Syria (After the Arab Spring)

4.2.1 Gender Dynamics and Family Structures

In post-Arab Spring Syria, the private sphere remains central to women's lives, where women's roles are simultaneously constrained and redefined. To understand how empowerment is negotiated in contemporary Syria, it is important to examine how these roles have shifted within the household, especially amid the weakening or absence of traditional male authority.

Despite years of upheaval, Syrian society continues to be shaped by deeply rooted patriarchal norms. With a Gender Inequality index of 0.55 (Degnan, 2024), concepts like family honor and kinship loyalty continue to shape social structures. Women are often perceived as bearers of family reputation, a role that subjects them to controls over their

behavior, mobility and public presence. These restrictions are particularly pronounced in rural areas and among displaced communities, where patriarchal traditions have been reinforced as mechanisms of stability amid broader social collapse (Evanson, 2016).

Family structures remain largely patrilineal, with male authority, typically the father or eldest son, governing financial decisions and legal rights. This is especially visible in legal guardianship over children and inheritance rights, where women have little decision-making power. For instance, even widows or divorced women often face legal and cultural barriers in transferring property or naming rights to their children, underscoring their marginal position within the household and society. Further, early and arranged marriages are still prevalent, especially in rural areas where economic hardship encourages families to marry off daughters at younger ages (IICISAR, 2023). While many women now access education, prevailing gender norms still prioritize marriage to a wealthy husband over academic achievement or professional development, weakening young women's awareness of their rights and long-term independence (Degnan, 2024).

However, the protracted Syrian conflict has profoundly reshaped household structures, forcing a reconfiguration of traditional gender roles and responsibilities. The mass displacement, imprisonment and death of male household members have pushed many women into positions of primary responsibility, becoming *de facto* HHs in a society where legal and cultural frameworks still overwhelmingly privilege male authority (Alsaba & Kapilashrami, 2016, p. 9). This shift has occurred not through deliberate empowerment policies, but because of survival in crisis, where women had to assume central roles in decision-making, income generation and caregiving. Statistical evidence reflects these shifts, showing that while only 4% of Syrian households were female-headed before 2011, this figure rose dramatically to 22% in the aftermath of the conflict (Cookman, 2021). Yet, these shifts in gender roles remain precarious. They are not backed by institutional reforms or legal protections and women continue to face systemic barriers to autonomy. Divorce, for instance – though legally sanctioned – carries significant social stigma, often resulting in women's exclusion from the labor market and social life, particularly in conservative or rural communities (Evanson, 2016).

These contradictions underscore the fragility and ambivalence of change. On the one hand, conflict has opened new spaces for women to exercise agency. On the other, without structural change, such agency remains vulnerable to reversal. The war has not dismantled

patriarchy – it has displaced it, forcing women to navigate new roles under old constraints. Their empowerment, therefore, must be understood not as a linear gain, but as a contested and uneven process, shaped as much by necessity as by resistance.

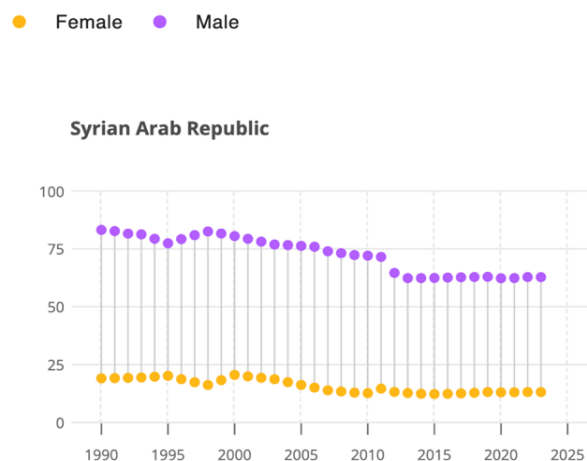
4.2.2 Women and Education in Syria

Before the conflict, Syria saw significant improvements in girls' access to education, with national literacy rates approaching 78% between 2005 and 2011 (Zeno, 2021, p. 285). However, the war has caused a reversal of these gains. According to UNICEF, one in three schools in Syria has been destroyed, damaged or repurposed for military use, severely limiting safe and accessible education (UNICEF, 2025). According to the German Federal Ministry for Economic Cooperation and Development, in 2021 over 2.5 million children inside Syria were out of school, 40% of whom were girls (BMZ, 2021). The security situation, coupled with long travel distances and economic hardship, has contributed to a sharp decline in girls' enrollment, especially in secondary and higher education (Zeno, 2021, p. 286). The state's investment in education also fell significantly: between 2010 and 2019, public expenditure on education declined by 71.5% (Zeno, 2021, p. 286). This led to Syria's rank in the Human Development Index for education going from 122nd in 2010 to 184th in 2019 (SCPR, 2020). While education remains officially free and compulsory for nine years, accessibility is largely unequal, with marginalized groups, especially displaced girls and rural women, bearing the brunt of exclusion.

4.2.3 Women and the Syrian Labor Market

The Syrian conflict has profoundly transformed the labor market, leading many women to assume primary economic responsibilities in response to the widespread death, detention or displacement of male relatives (Cookman, 2021). However, Syrian women entered the labor market with limited experience, minimal access to resources, and in most cases, without institutional support to protect them from exploitation, harassment or legal exclusion (Cookman, 2021).

Figure 2. Labor Force Participation Rate by Sex in the Syrian Arab Republic (% of population ages 15+), 1990–2023



Source: World Bank (2024). *Labor force participation rate, by sex (% of population ages 15+) – Modeled ILO estimate*. World Development Indicators. Retrieved from <https://data.worldbank.org>

They predominantly work in the informal sector, where they take on low-paid, home-based roles such as sewing, food processing, hairdressing and handicrafts. These jobs often mirror traditional domestic responsibilities, reflecting gender norms that shape the division of labor. In addition to this, access to formal employment is limited due to persistent structural barriers, including mobility restrictions, inadequate childcare, gender stereotypes and legal discrimination (Hijazi, 2022). One of the most visible causes of these constraints is women's continued underrepresentation in decision-making and leadership roles, which limits their capacity to influence labor policy or advocate for protections against workplace abuse and discrimination. Despite a growing recognition

of women as economic contributors deep-rooted patriarchal norms continue to restrict female labor force participation (Cookman, 2021).

This is reflected in recent labor statistics: as of 2023, only 13.4% of women in Syria were active in the labor force, compared to 63.1% of men (World Bank, 2024), underscoring the persistence of gendered inequalities in both formal employment and economic decision-making. In response to these challenges, various post-2018 economic empowerment programs have sought to support women, particularly widows and FHHs, through grants and livelihood initiatives. While such interventions are crucial in humanitarian contexts, they have largely operated within a vulnerability framework, treating women as beneficiaries rather than agents of change. By prioritizing immediate survival over long-term autonomy, these programs often reinforce traditional caregiving roles and overlook the broader structural reforms necessary to expand women's agency. Failing to address women's unpaid labor or to challenge the broader structural inequalities that constrain their agency, such programs risk entrenching existing gender hierarchies (Hijazi, 2022).

5. Research Aims and Hypotheses

Building on the preceding theoretical and contextual analysis, this study addresses a key gap in understanding how gendered power dynamics shape household food security in conflict-affected settings. While existing research affirms the role of women's empowerment in improving food security outcomes (Essilfie et al., 2024; Clement et al., 2019), less attention has been paid to *marital status* as a structural determinant of women's agency, resource access and social legitimacy (Kabeer, 1999). In the Syrian context this study explores how marriage and widowhood differently influence women's ability to ensure household food security.

Rather than treating female headship as a proxy for empowerment, this research interrogates how the three pillars of this process are influenced by women's marital status in fragile settings. It does so through a series of interrelated hypotheses examining both direct and moderated relationships between empowerment resources and food security:

H1: Widowed women are more likely to experience household food insecurity than married women. This reflects the idea that, in patriarchal societies, widowhood entails the loss of male-mediated access to property, legitimacy and social protection – leaving women structurally disadvantaged in securing food for their households.

H2a: Land ownership is associated with reduced household food insecurity. Drawing on literature linking asset control to improved well-being, this hypothesis views land as a vital empowerment resource, particularly in agrarian economies like Syria.

H2b: The food security benefits of land ownership are stronger for widowed women than for married women. In the absence of a spouse, land may serve as a compensatory source of empowerment, granting widows greater agricultural autonomy and enabling them to offset structural exclusions.

H3a: Women’s education is associated with reduced household food insecurity. Based on existing literature, education enhances women’s capacity to access information, earn income, and make strategic decisions about food provisioning, thereby improving household food security.

H3b: The food security benefits of education are stronger for widowed women than for married women. In the absence of a spouse, education may become a particularly critical resource for widows’ empowerment, enabling them to overcome institutional and social barriers.

6. Methodology and Data

6.1 Data Collection

This study is based on primary household-level data collected from 5 Syrian governorates (Hama, Deir-ez-zor, Aleppo, Homs and Rural Damascus) in the Syrian Arab Republic. The data collection focused on agricultural households and was conducted in two waves: March 2023 and March 2024. Both rounds were implemented just before the Ramadan

period to minimize any seasonal or religious biases related to food consumption and dietary behavior. The dataset consists of a balanced panel of 466 households tracked over time, drawn from a larger initial cross-sectional survey of 1,733 households distributed across the governorates. All interviews were conducted at the household level, with responses provided by the HH. The survey instrument is a multi-module questionnaire designed to assess livelihood, food security and gender-related indicators. It includes 16 thematic modules covering household location and demographics, access to services, assets, agricultural activities, social safety nets, food security, coping strategies, value chain engagement, information access and gender roles in decision-making and empowerment.

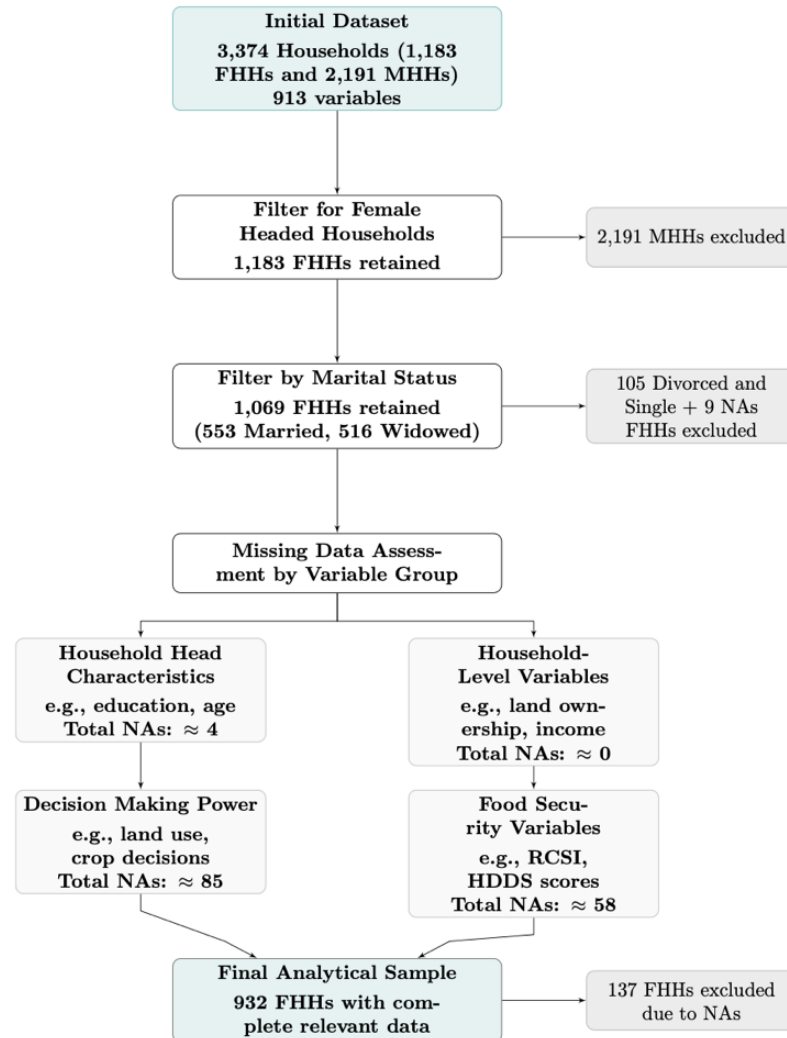
Data collection was conducted through in-person, paper-based interviews by a team of 44 trained enumerators. A centralized team oversaw the manual data entry using Microsoft Access to ensure accuracy and consistency in reporting. Following data collection, a thorough cleaning process was performed. Irregularities – such as geographic inconsistencies or mismatches between sampling design and responses – were corrected prior to analysis. All data was collected ethically, with verbal informed consent obtained from all participants. Households were assured of confidentiality and their right to withdraw from the study at any time. Each household was assigned an anonymized ID for matching across survey waves and to protect personal data.

6.2 Analytical Sample

The analytical sample for this study was drawn from an anonymized reduced dataset provided by ISDC, which comprises two waves of panel data collected at the household level. Given the focus of this research on women's empowerment and the potential influence of marital status on resilience and food security outcomes, the provided dataset was restricted to FHHs. From the full sample of 3,374 households, only FHHs (N = 1,183) were retained. To maintain analytical consistency and ensure meaningful group comparisons, the sample was further restricted to female heads who were either married or widowed (N = 1,069), excluding those who were divorced or single due to their small

representation in the dataset ($N = 105$). Restricting the sample in this way enabled a more focused analysis of the relationship between women’s empowerment and household food security, while accounting for the influence of marital status as a key determinant of empowerment. A subset of variables – spanning characteristics, land ownership, empowerment indicators and food security outcomes – was retained. Observations with missing data in any of these domains were excluded: this included 4 cases with missing sociodemographic data, 58 with incomplete food security scores and 85 missing decision-making power indicators. After removing these incomplete cases, the final analytical sample consisted of **932 FHHs** with complete and relevant data, as shown in Figure 3.

Figure 3. presents the flow of sample selection, from the full dataset to the final analytical sample.



6.3 Variables

6.3.1 Outcome Variables: Household Food Security

While the concept of food security is globally recognized (FAO, 1996), its measurement at the household level presents significant challenges (Jemaneh & Shibeshi, 2023). Due to its multidimensional nature, food security cannot be effectively captured through a single indicator (Jemaneh & Shibeshi, 2023; WFP, 2008). Instead, robust analyses must account for the four core pillars of food security: availability, access, utilization (nutrition) and stability over time (Essilfie et al., 2024, p. 3). For this reason, to capture this complexity, researchers have developed a wide range of internationally accepted indicators (Manikas, Ali, & Sundarakani, 2023).

The selection of indicators in any given study typically depends on the research objectives, data availability and practical constraints (Jemaneh & Shibeshi, 2023). This study draws on a dataset that offers a rich array of variables that capture not only the quantity and diversity of food consumed but also behavioral responses to food shortages. A multidimensional approach combining the FCS, HDDS and RCSI was adopted to avoid the misclassification risks of relying on a single metric (Maxwell, Vaitla, & Coates, 2013; Manikas, Ali, & Sundarakani, 2023). These combined food security and insecurity indicators are crucial in conflict-affected settings like Syria, where food access and availability are volatile and households must adapt to prolonged instability (Doocy et al., 2017). Using HDDS, FCS and RCSI together provides a more comprehensive view of immediate food security status and households' adaptive capacities, making them well-suited for sensitive environments (Doocy et al., 2017). Moreover, since this research uses marital status as a proxy for women's empowerment, applying a multi-dimensional measurement approach is not only methodologically robust but also critical for providing a more nuanced understanding of how structural vulnerabilities affect women's ability to secure household food access at different levels. This enhances the reliability of findings and ensures a fuller understanding of the food security realities faced by FHHs.

The Household Dietary Diversity Score

The HDDS serves as a proxy for household food access, dietary quality and diversity (Galiè et al., 2019; Sraboni et al., 2014). Greater dietary diversity is strongly associated with improved child growth, higher birth weight, better overall nutritional status and increased energy and protein intake (USAID, 2006). In addition, by measuring a household's ability to access a nutritionally diverse and adequate diet, the HDDS also offers insight into the household's socio-economic status (Kennedy, Ballard, & Dop, 2011).

The HDDS measures the number of food groups consumed by household members over the past 24 hours across 12 categories: cereals, roots/tubers, vegetables, fruits, meat, poultry/offal, eggs, fish/seafood, pulses/legumes/nuts, milk/milk products, oils/fats, sugar/honey and miscellaneous foods (Equation 1) (USAID, 2006)¹. However, the HDDS has not been validated against a standard of adequacy and no universal cut-off exists to distinguish sufficiently diverse diets from insufficiently diverse ones at the household level (Data4Diets, 2023). In this study, HDDS analysis helps examine how marital status affects resource access and the ability to maintain a balanced household diet.

The Food Consumption Score

The FCS, developed by the World Food Programme (WFP, 2019), has become one of the most widely used proxy indicators for household food security and measures both the quantity and quality of food consumed over the past seven days (Manikas, Ali, & Sundarakani, 2023, p. 16; Jemaneh & Shibeshi, 2023). It differs from the HDDS by including consumption frequency and food group nutritional weighting.

Following WFP guidelines (2024), the FCS is calculated by multiplying the number of days each of eight standardized food groups was consumed by its assigned weight, then summing the results into a composite score (Equation 2). Households are categorized into

¹ This set of food groups is derived from the U.N. Food and Agriculture Organization (Food and Agricultural Organization. Food Composition Table for Africa. Rome, Italy, 1970. As viewed at www.fao.org/docrep/003/X6877E/X6877E00.htm.

food security levels based on thresholds, adjusted here for Syria’s rich and diverse diet (WFP, 2024).

Figure 4 FCS Thresholds

Food consumption group	Standard thresholds	Adjusted thresholds ⁴
Poor food consumption	0–21	0–28
Borderline food consumption	21.5–35	28.5–42
Acceptable food consumption	35.5–112	42.5–112

The reduced Coping Strategy Index

The RCSI measures household food insecurity by capturing the frequency and severity of coping behaviors during food shortages, making it particularly valuable in conflict settings (Maxwell & Caldwell, 2008). It evaluates five coping strategies:

1. Relying on less preferred or cheaper foods
2. Borrowing food or relying on help from friends or relatives
3. Limiting portion sizes at mealtime
4. Restricting consumption by adults so that small children can eat
5. Reducing the number of meals eaten in a day (Maxwell & Caldwell, 2008).

Each strategy has a universal severity weight and frequency responses over the past seven days are multiplied by these weights and summed into a composite RCSI score (Equation 3). Higher scores indicate more severe food insecurity. Although no universal threshold exists, RCSI is often treated as a continuous variable or adapted to local classifications (Maxwell & Caldwell, 2008; WFP, 2008). Unlike static indicators, the RCSI captures short-term adaptive responses, offering critical insights into both household stress and resilience. Together with the FCS and HDDS, it provides a multidimensional assessment linking food consumption patterns with households’ adaptive responses (Maxwell, Vaitla, & Coates, 2014; WFP, 2008).

6.3.2 Independent Variable: Marital Status

The primary independent variable in this study is the marital status of the female household head, operationalized as a proxy for intra-household power dynamics and women's differential access to resources in patriarchal settings. Respondents were asked, "*What is the marital status of the HH?*" selecting among married, divorced, single or widowed. Following data inspection, divorced and single female heads – representing only 6.4% of the sample – were excluded to ensure statistical robustness. The final binary variable distinguishes between married and widowed women, with marital status retained as a factor variable.

While not used as a direct proxy for empowerment, marital status in this study functions as a key stratifying variable to examine how structural conditions – such as widowhood – shape women's access to empowerment resources like land and education. Measuring women's empowerment is inherently complex and often requires multidimensional data on agency, control over resources, mobility and gender norms (Malapit et al., 2015), which are frequently unavailable in conflict-affected settings. In light of these data constraints, this study focuses on land ownership and education as observable empowerment-related variables that reflect women's access to material and cognitive resources central to household food security.

6.3.3 Control Variables

Several control variables are included to isolate the relationship between marital status and household food security outcomes. Their selection is based on theoretical considerations and preliminary data analysis.

Age of the HH plays a complex role: older women may possess greater authority and decision-making power (Essilfie et al., 2024, p. 4), while economic vulnerability could

persist. In the analysis, age is included as a continuous variable without log transformation, based on the distribution's normality.

Education enhances women's capacity for resource management, nutritional awareness and economic participation (Adepoju et al., 2015, p. 105). Although preliminary correlation analyses in this study suggest a weak relationship between education and food security outcomes, education remains included as a control variable for theoretical completeness and comparability with previous research.

Access to land is a major factor enabling women's empowerment and food security, especially in agrarian economies (Quisumbing et al., 1996, p. 1). Here, **Total Household Owned Land Size** was calculated by summing irrigated and rainfed areas in donum², with a log transformation applied to correct for skewness. Land ownership is included as a structural variable influencing women's capacity to ensure household food security.

Humanitarian aid directly impacts household food security by providing material support that mitigates food shortages (Mataka et al., 2023, p.5). In this study, a binary variable was created to capture whether the household received any form of aid. A household was coded as having received aid, if it reported receiving either regular formal cash transfers, informal cash transfers or in-kind assistance from United Nations agencies, government bodies or non-governmental organizations; otherwise, it was coded as 0. This was operationalized by summing binary indicators for each aid type and assigning a value of 1 if any assistance was reported. Controlling for aid access is crucial to avoid misattributing food security outcomes solely to household characteristics when external support plays a role.

² A donum (or dunum) is a traditional unit of land measurement in Syria, equivalent to 1,000 square meters (approximately 0.1 hectares or 0.247 acres).

6.4 Methodology Strategy

To examine the relationship between marital status and household food security in Syria, this study employs both descriptive and econometric methods. Given data limitations – including a relatively small sample size and lack of panel structure – more sophisticated econometric techniques were not feasible. As a result, the analysis relies on a pooled OLS regression model, which treats each observation as independent and does not account for unobserved, time-invariant household characteristics. This may introduce bias, particularly from omitted variables that do not vary over time.

Despite these limitations, pooled OLS offers a useful exploratory framework to examine associations between women’s marital status and household food security. Standard errors are clustered at the household (ID) level to improve the reliability of estimates and partially account for intra-household correlation across survey waves.

The analysis consists of nine regressions across three food security indicators, with three models for each:

- I. Base Model: Examines the direct relationship between marital status and food security while controlling for land ownership and education effects.

$$\text{FoodSecurity}_i = \beta_0 + \beta_1 \cdot \text{MaritalStatus}_i + \beta_2 \cdot \text{Governorate}_i + \beta_3 \cdot \text{Education}_i + \beta_4 \cdot \text{Age}_i + \beta_5 \cdot \log(\text{LandOwned}_i) + \beta_6 \cdot \text{AnyAid}_i + \varepsilon_i \quad (4)$$

- II. Land Interaction Model: Adds an interaction between marital status and land ownership to test whether widowed FHHs benefit differently from land resources compared to married FHHs.

$$\text{FoodSecurity}_i = \beta_0 + \beta_1 \cdot \text{MaritalStatus}_i + \beta_2 \cdot \log(\text{LandOwned}_i) + \beta_3 \cdot (\text{MaritalStatus}_i \times \log(\text{LandOwned}_i)) + \beta_4 \cdot \text{Governorate}_i + \beta_5 \cdot \text{Education}_i + \beta_6 \cdot \text{Age}_i + \beta_7 \cdot \text{AnyAid}_i + \varepsilon_i \quad (5)$$

- III. Education Interaction Model: Incorporates an interaction between marital status and education to test whether widowed FHHs benefit differently from education compared to married FHHs.

$$\text{FoodSecurity}_i = \beta_0 + \beta_1 \cdot \text{MaritalStatus}_i + \beta_2 \cdot \text{Education}_i + \beta_3 \cdot (\text{MaritalStatus}_i \times \text{Education}_i) + \beta_4 \cdot \text{Governorate}_i + \beta_5 \cdot \text{Age}_i + \beta_6 \cdot \log(\text{LandOwned}_i) + \beta_7 \cdot \text{AnyAid}_i + \varepsilon_i \quad (6)$$

This approach allows to investigate whether widowhood increases food insecurity while testing if empowerment resources like land and education function as compensatory mechanisms for women without male-mediated access to resources.

7. Descriptive Analysis

Before proceeding to multivariate analyses, a descriptive analysis was conducted to provide a comprehensive overview of the household sample. This initial step allows for a better understanding of the key characteristics of FHHs, their socio-demographic profiles and their food security outcomes.

7.1 Sample Characteristics

Table 1 presents key demographic and socioeconomic characteristics of the 932 FHHs included in the study, disaggregated by marital status. The sample comprises 483 married and 449 widowed HHs, offering a comparative view of the two groups most relevant to this analysis.

The sample is relatively well-distributed across the five governorates. Married women are most represented in Homs (33.5%) and Aleppo (31.5%), while widowed women are more concentrated in Hama (28.5%). Rural Damascus accounts for the smallest share overall (7.3%), reflecting possible spatial variations in displacement or aid distribution. Widowed women are substantially older on average (mean = 50.0 years) than married women (mean = 42.1 years), suggesting different life stages and possibly different experiences of conflict-related loss or transition.

Educational disparities between the two groups are pronounced: over half of widowed women (50.8%) report no formal education, compared to 30.4% of married women. Married women are nearly twice as likely to have higher education (19% vs. 8%), indicating generational and structural gaps in access to learning. These differences suggest that younger, married women may have benefited more from Syria's pre-conflict educational reforms having more educational opportunities (Zeno, 2021, p. 285). Additionally, married FHHs report slightly higher logged land ownership (mean = 1.24) than widowed FHHs (mean = 1.08), highlighting the influence of marital status in access to agricultural resources, still both groups exhibit modest landholdings overall.

Receipt of economic aid is high across both groups, with nearly all households reporting some form of assistance (mean $\approx$ 0.96). This underscores the widespread reliance on external support among FHHs, regardless of marital status.

Table 1. Summary Statistics Syrian Households

	Married (N=483)	Widowed (N=449)	Overall (N=932)
Governorate of Residence			
Aleppo	152 (31.5%)	90 (20.0%)	242 (26.0%)
Deir-ez-Zor	69 (14.3%)	65 (14.5%)	134 (14.4%)
Hama	74 (15.3%)	128 (28.5%)	202 (21.7%)
Homs	162 (33.5%)	124 (27.6%)	286 (30.7%)
Rural Damascus	26 (5.4%)	42 (9.4%)	68 (7.3%)
Age of Household Head			
Mean (SD)	42.1 (11.0)	49.9 (12.7)	45.8 (12.5)
Median [Min, Max]	41.0 [20.0, 85.0]	50.0 [18.0, 93.0]	45.0 [18.0, 93.0]
Educational Attainment			
No Education	147 (30.4%)	228 (50.8%)	375 (40.2%)
Primary Education	244 (50.5%)	185 (41.2%)	429 (46.0%)
Higher Education	92 (19.0%)	36 (8.0%)	128 (13.7%)
Total Land Owned (Logged)			
Mean (SD)	1.24 (1.04)	1.08 (0.977)	1.16 (1.01)
Median [Min, Max]	1.39 [0, 3.43]	1.10 [0, 3.47]	1.10 [0, 3.47]
Receipt of Economic Aid			
Mean (SD)	0.965 (0.184)	0.958 (0.202)	0.961 (0.193)
Median [Min, Max]	1.00 [0, 1.00]	1.00 [0, 1.00]	1.00 [0, 1.00]

7.2 Household Food Security across Married and Widowed Women

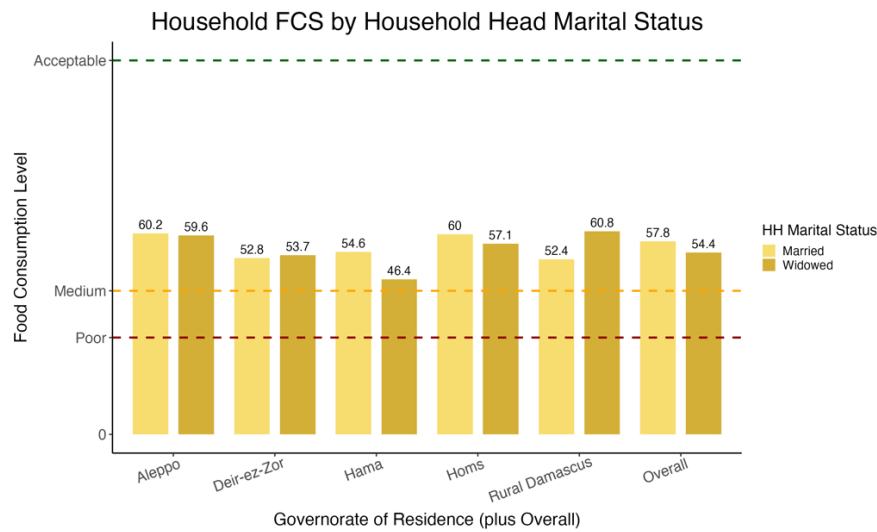
The Household Food Consumption Score

The FCS categorizes households as poor (0–28), medium (29–42.5) or acceptable (42.6+). According to Figure 5, the average score lies within the acceptable threshold, suggesting a basic/good level of food security. This reinforces the limitations of relying on a single indicator, especially in a resource-rich like Syria, where the FCS alone may mask underlying food insecurity (Almukaddem et al., 2022). However, values remain low and close to the medium threshold, highlighting persistent vulnerability in diet quality and quantity (FAO, 2021).

Marital status differences are present: married FHHs generally report slightly higher or comparable scores to widowed ones. Regional trends reveal further variation. In Aleppo, Homs and Rural Damascus, both groups score around or above 60, indicating relatively better food access. Conversely, Deir-ez-Zor and Hama show lower levels, with widowed women in Hama averaging 46.4, just above the medium threshold, suggesting heightened vulnerability.

Looking across governorates, slight score differences between widowed and married FHHs reflect the impact of male absence on resource access. Widowed heads, lacking spousal support, face greater difficulty providing adequate diets, reinforcing their structural disadvantage in ensuring food security. These findings underscore how both conflict exposure and local conditions shape food outcomes, particularly affecting widowed FHHs.

Figure 5. Household FCS average by Governorate – baseline and midline data combined



The Household Dietary Score

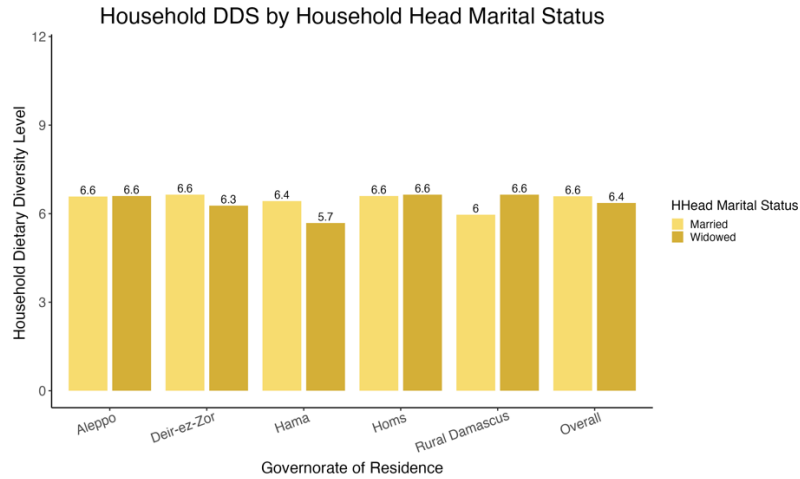
As shown in Figure 6, HDDS levels in this sample are modest across all regions, with most values clustering around 6.6, with little variation between married and widowed female heads. This uniformity suggests that FHHs consistently face significant barriers to accessing diverse diets, likely reflecting the compounded effects of conflict, economic collapse and structural gender inequalities that affect women regardless of their marital status.

Nevertheless, regional disparities exist. In Hama, widowed women report a notably lower HDDS (5.7) compared to married heads (6.4), pointing to greater nutritional vulnerability. Conversely, in Rural Damascus, widowed women display slightly higher HDDS scores than married women (6.6 vs. 6.0), indicating greater resilience among widows in this region – a pattern also observed for their FCS. These findings suggest that while widowhood generally signals vulnerability, the interaction between gender norms, local conflict dynamics and humanitarian support may sometimes enable widowed women to secure better food access compared to married women, particularly in specific localized contexts.

Overall, the modest dietary diversity observed across the sample highlights the persistent structural disadvantages facing Syrian FHHs in conflict-affected areas, revealing that

household headship status alone is insufficient to overcome entrenched inequalities in food security.

Figure 6. HDDS - baseline and midline data combined

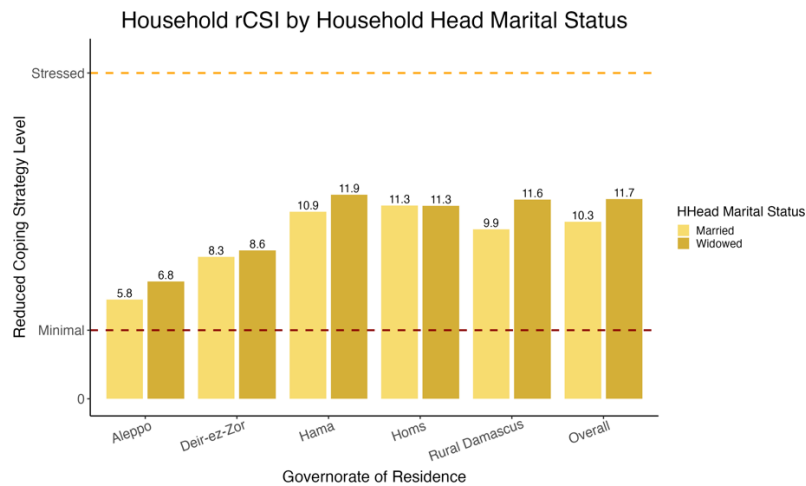


The Household Reduced Coping Strategy

As illustrated in Figure 7, RCSI levels in this sample show notable variation across regions and between married and widowed FHHs. Overall, widowed female heads report slightly higher RCSI scores than married heads (11.7 vs 10.3), suggesting that widows rely more heavily on negative coping mechanisms to manage food shortages. In governorates like Hama and Rural Damascus, widowed households show particularly high RCSI values (11.9 and 11.6, respectively), approaching the “stressed” coping threshold.

These findings underscore that widowhood often entails not only a loss of male-associated economic support but also increased vulnerability to food insecurity. Despite being HHs, widowed women in Syria face structural and economic barriers that intensify reliance on negative coping strategies. The geographic disparities further reflect how localized conflict severity, displacement and humanitarian access continue to shape households’ ability to cope.

Figure 7. Household Reduced Coping Strategy Index - baseline and midline data combined



7.4 Women's Decision-Making Power over Agricultural Resources

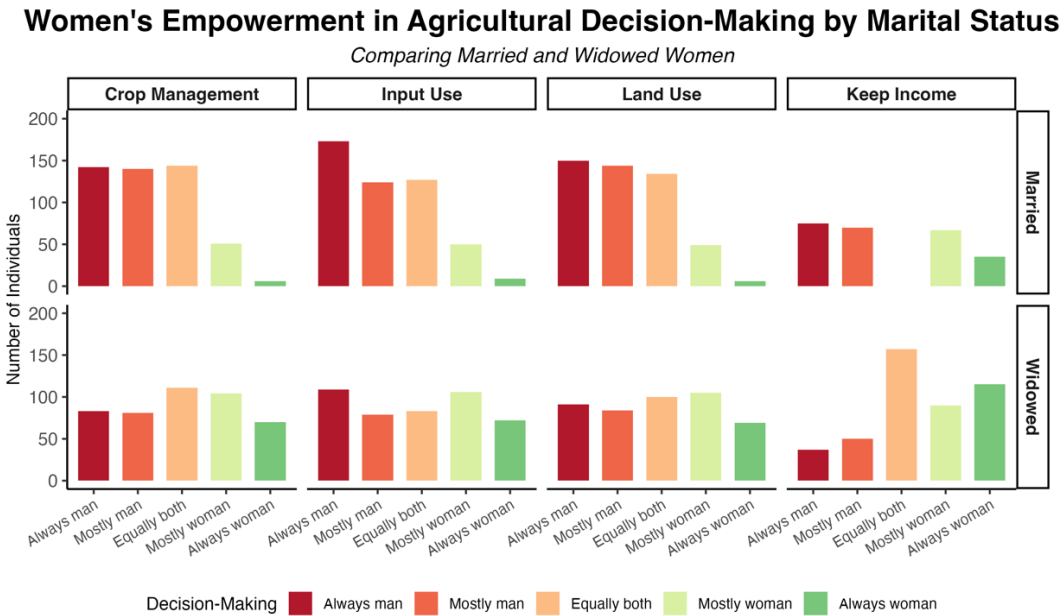
As previously stated, empowerment is best understood as a dynamic process involving three interrelated dimensions (Kabeer, 1999). In patriarchal societies like Syria, marital status becomes a critical determinant of this process, shaping women's ability to access resources, their capacity to exercise independent decision-making and their potential to achieve meaningful outcomes such as sustaining household food security (Van de Walle, 2011).

In such settings, empowerment is not simply about formal authority but about navigating legal, social and economic barriers (Evanson, 2016; Dercon & Krishnan, 2000). Moreover, women's status and rights are largely mediated through their association with men, particularly husbands (Evanson, 2016). As a result, while married women may enjoy greater access to resources through their male partners, their decision-making authority is significantly constrained. Therefore, it is also crucial to analyze how marital status affects female heads' agency.

To illustrate this dynamic, Figure 8 compares married and widowed women's decision-making power across four agricultural domains: crop management, input use, land use and income recipient from agricultural production. The results show a clear pattern: married women overwhelmingly report lower independent decision-making across all

domains, with decisions largely dominated by male authority. Even in managing agricultural income, most married women share decision-making with their husbands undermining their economic autonomy.

Figure 8. Women's Empowerment in Agricultural Decision-Making by Marital Status: Comparing Married and Widowed Women



Widowed women, by contrast, report higher levels of independent decision-making, particularly over land use and agricultural income. However, as earlier sections demonstrated, this apparent gain in agency does not translate into improved household food security. Structural barriers continue to undermine widowed women's capacity to sustain household welfare. Additionally, in some cases their authority is undermined by male relatives who exert control over household decisions, reflecting the persistent influence of patriarchal norms even in the absence of a spouse.

8. Findings

8.1 Head Marital Status and Household food Security

The first stage of the analysis investigates the relationship between marital status and household food security among FHHs in Syria. Following Chapter 2 that positions marital status as a structural determinant of women's empowerment in patriarchal contexts, this model tests *H1*: widowed women are more likely to experience household food insecurity than married women.

Table 2 presents the pooled OLS regression results, which offer strong empirical support for *H1*. The most pronounced effect is observed in the FCS model, where widowed women have, on average, 3.4 point lower FCS compared to married women ($\beta = -3.4$, $p = 0.004$), indicating a substantively meaningful reduction in the ability to maintain a consistent and nutritionally adequate diet. A similar pattern appears in the HDDS model, where widowhood is associated with a 0.22 point decrease in dietary diversity scores ($\beta = -0.22$, $p = 0.079$). Although this association is only marginally significant, it suggests that widowed women face greater obstacles in accessing diverse food sources, likely due to limited financial resources and exclusion from production or market systems. This disparity is further reflected in the RCSI, where widowed women have, on average, 0.72 point higher coping strategy scores compared to married women ($\beta = 0.72$, $p = 0.071$). While this may indicate heightened vulnerability in crisis times, it could also reflect adaptive behaviors developed in response to structural exclusion, enabling these women to mitigate acute food shortages.

Together, these findings offer robust support for *H1*, confirming that marital status significantly shapes women's ability to secure household food security. While the overall sample scores approach acceptable food security thresholds (Figures 5–7), these averages obscure sharp disparities linked to marital status. Married women are structurally better positioned to secure household food needs, while widowed women, despite their greater household autonomy, lack access to the resources necessary to translate that power into improved food outcomes. Finally, regional variation across all three indicators further

underscores the role of localized conflict dynamics, highlighting the need to account for regional context when analyzing food security in fragile settings.

Table 2. Pooled OLS Regression Summary Table for each Household Food Security and Insecurity Indicators

RCSI				FCS				HDDS			
Characteristic	Beta	95% CI	p-value	Characteristic	Beta	95% CI	p-value	Characteristic	Beta	95% CI	p-value
Marital Status of Household Head				Marital Status of Household Head				Marital Status of Household Head			
Married	—	—		Married	—	—		Married	—	—	
Widowed	0.72	-0.06, 1.5	0.071	Widowed	-3.4	-5.7, -1.1	0.004	Widowed	-0.22	-0.47, 0.03	0.079
Governorate of Residence				Governorate of Residence				Governorate of Residence			
Aleppo	—	—		Aleppo	—	—		Aleppo	—	—	
Deir-ez-Zor	2.2	0.98, 3.4	<0.001	Deir-ez-Zor	-4.8	-8.4, -1.1	0.010	Deir-ez-Zor	0.23	-0.16, 0.62	0.3
Hama	6.7	5.5, 7.9	<0.001	Hama	-9.2	-13, -5.6	<0.001	Hama	-0.37	-0.75, 0.01	0.054
Homs	7.9	6.8, 9.0	<0.001	Homs	0.33	-3.0, 3.7	0.8	Homs	0.18	-0.17, 0.54	0.3
Rural Damascus	6.0	4.3, 7.6	<0.001	Rural Damascus	0.75	-4.0, 5.5	0.8	Rural Damascus	-0.01	-0.53, 0.50	>0.9
Education Level of Household Head				Education Level of Household Head				Education Level of Household Head			
No Education	—	—		No Education	—	—		No Education	—	—	
Primary Education	-0.21	-1.2, 0.75	0.7	Primary Education	-1.3	-4.2, 1.5	0.4	Primary Education	-0.08	-0.38, 0.22	0.6
Higher Education	-0.02	-1.3, 1.3	>0.9	Higher Education	-3.3	-7.1, 0.62	0.10	Higher Education	-0.20	-0.61, 0.22	0.4
Age of Household Head	-0.02	-0.05, 0.02	0.4	Age of Household Head	0.13	0.03, 0.23	0.014	Age of Household Head	0.01	0.00, 0.02	0.2
Log Household Land Size Owned	-0.44	-0.81, -0.07	0.020	Log Household Land Size Owned	1.1	0.04, 2.2	0.043	Log Household Land Size Owned	0.14	0.03, 0.26	0.017
Receipt of Economic Aid	2.2	0.22, 4.2	0.029	Receipt of Economic Aid	11	5.3, 17	<0.001	Receipt of Economic Aid	0.37	-0.26, 1.0	0.2
Abbreviation: CI = Confidence Interval				Abbreviation: CI = Confidence Interval				Abbreviation: CI = Confidence Interval			

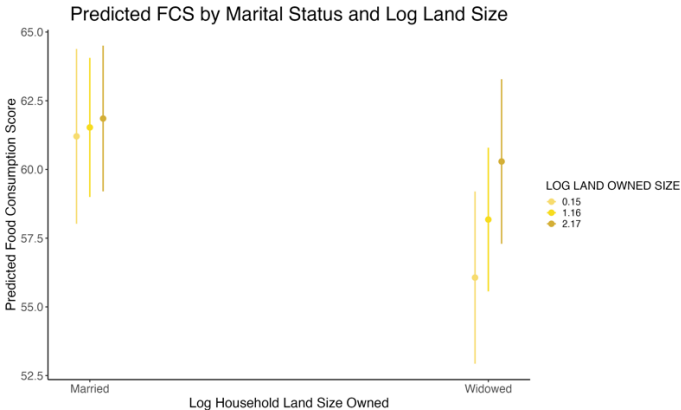
8.2 Household Land Ownership, Marital Status and Household food Security

The second stage of the analysis investigates the role of land ownership in shaping household food security and whether its effects vary by marital status. Drawing on literature linking asset control to improved household wellbeing (Quisumbing et al., 1996, p. 1), *H2a* posits that land ownership reduces household food insecurity.

Initial results from Table 2 support *H2a*: each log-unit increase in land size is associated with a significant 0.44 point decrease in food-related coping strategies (RCSI, $\beta = -0.44$, $p = 0.020$) and a 0.14 point increase in dietary diversity (HDDS, $\beta = 0.14$, $p = 0.017$) and a 1.1 point increase in food consumption (FCS, $\beta = 1.1$, $p = 0.043$). These results are reinforced in Table 3, where RCSI drops by 1.2 points ($p < 0.001$), confirming land's protective role. In Tables 4 and 5, the positive direction holds, though statistical significance is not maintained. Taken together, these findings provide partial support for *H2a*, suggesting that land ownership is generally associated with improved household food security, particularly in reducing reliance on coping strategies.

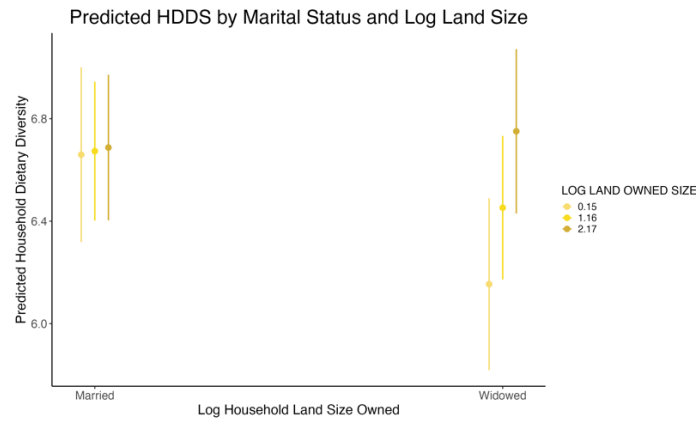
To test *H2b*, Model II introduces interaction terms between marital status and logged household land size (Table 3-5). This assesses whether land compensates for widowed women’s structural disadvantages. In the FCS model (Table 4, Figure 9), the interaction suggests on average that each log-unit of land adds 1.8 points to widowed women’s FCS relative to married women, though not statistically significant ($\beta = 1.8$, $p = 0.10$ in Model 2.5). Still, Figure 9 shows a steeper FCS rise for widows, with a 3.6 point advantage at higher land levels. Although wide confidence intervals preclude definitive conclusions, the trend suggests that land ownership may help mitigate some of the structural disadvantages widowed women face in achieving adequate household food consumption.

Figure 9. Predicted FCS by Marital Status and Log Land Size



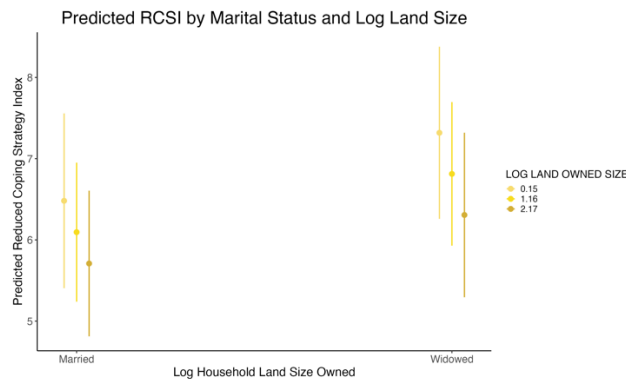
For the HDDS model, the interaction between widowhood and land size is both positive and statistically significant and unchanged across all models (Table 5), indicating that each log-unit increase in land size is associated with an additional 0.28 point increase in dietary diversity for widowed women compared to married women ($\beta = 0.28$, $p = 0.015$ in Model 3.5). Figure 10 illustrates this clearly: dietary diversity rises more sharply with land size for widowed households, offering strong support for *H2b*. This points to land as an especially meaningful empowerment resource where institutional support is limited or absent.

Figure 10. Predicted HDDS by Marital Status and Log Land Size



Regarding the RCSI model (Table 3, Figure 11), almost direction consistent over the 5 models, the interaction term between widowhood and land size is small (around 0.10 point decrease) and not statistically significant ($p = 0.70$). Figure 11 shows similar reductions in RCSI scores across both groups, suggesting that while land ownership reduces reliance on coping strategies, this effect does not differ substantially by marital status.

Figure 11. Predicted RCSI by Marital Status and Log Land Size



Overall, the interaction models offer partial support for *H2b*. While land ownership appears to benefit all households, widowed women gain more in terms of dietary diversity. This challenges the assumption that widows uniformly lack the capacity to convert resources into wellbeing. Instead, the absence of a male partner may create greater space for autonomous decision-making over land use. When widows do gain access to

land, they may be more likely to leverage it directly for household food needs. These findings align with broader research on adaptive agency in fragile settings (Clement et al., 2019; Essilfie et al., 2024), suggesting that under certain structural constraints, widowed women actively repurpose empowerment resources into meaningful improvements in household welfare. This is particularly visible in dietary diversity outcomes, which likely reflect a combination of subsistence cultivation and increased control over food procurement decisions.

8.3 Head Education Level, Marital Status and Household food Security

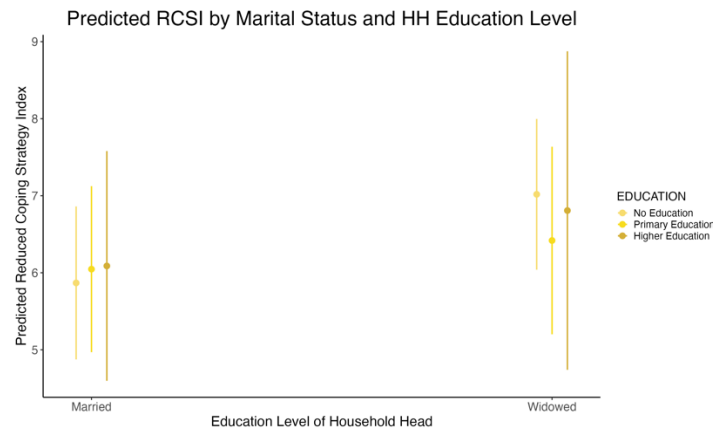
The third stage of analysis explores whether education improves household food security and if its effects differ by marital status. Based on prior evidence linking women's education to better food outcomes (Adepoju et al., 2015), *H3a* posits that higher education reduces household food insecurity.

Initial results from Table 2 offer limited support for *H3a*. On average, neither primary nor higher education is associated with statistically significant changes in RCSI ($\beta = -0.21$, $p = 0.7$; $\beta = -0.02$, $p > 0.9$), FCS ($\beta = -1.3$, $p = 0.4$; $\beta = -3.3$, $p = 0.1$) or HDDS ($\beta = -0.08$, $p = 0.6$; $\beta = -0.20$, $p = 0.4$), showing a counterintuitive trend: higher education is linked to slightly worse food outcomes, suggesting that in this conflict-affected setting, education alone may not offset broader gendered or structural constraints.

To test *H3b*, interaction models assess whether the effect of education differs by marital status (Tables 6–8). In the RCSI models (Table 6), education is initially associated with significant increase in coping strategy use: on average, a 3.3-point increase for primary education ($p < 0.001$) and a 4.4 point increase for higher education ($p < 0.001$) in Model 1.1. However, these effects, which become negative, lose significance once controls are added. Interaction terms remain non-significant (Model 1.5: $\beta = -0.78$, $p = 0.3$; $\beta = -0.43$, $p = 0.7$), indicating no added benefit for widows.

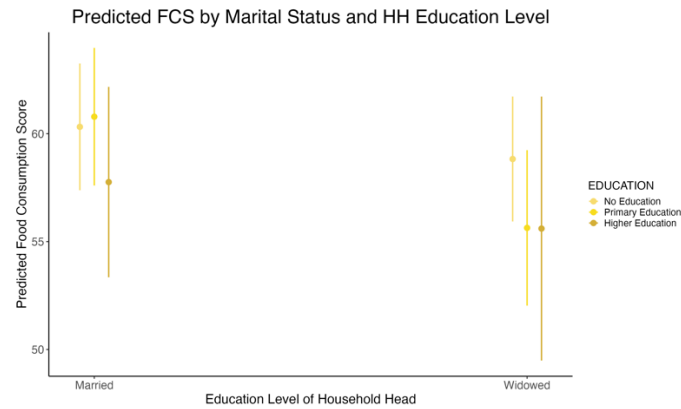
Figure 12 shows RCSI scores are higher for widows at all education levels and education does not appear to reduce this gap. In fact, RCSI slightly increases with education, though large confidence intervals limit the analysis.

Figure 12. Predicted RCSI by Marital Status and Education Level



In the FCS model (Table 7), education again shows no consistent positive effect for widowed FHHs. While higher education is initially associated with a significant 4.9-point decrease in FCS in Model 2.1 ($p = 0.032$), this effect loses significance in subsequent models (Models 2.2-2.5), suggesting that the relationship is likely confounded by other factors such as regional context or asset ownership. Primary education is never significant as well as the interaction terms which remain non-significant throughout (Model 2.5: $\beta = -3.7$, $p = 0.13$; $\beta = -0.66$, $p = 0.9$), indicating no compensatory effect. Figure 13 illustrates that FCS scores remain lower among widowed households across all education levels, with minimal within-group variation. Contrary to expectations, these results suggest that education may not improve FCS outcomes for widowed households in Syria and may even reduce them.

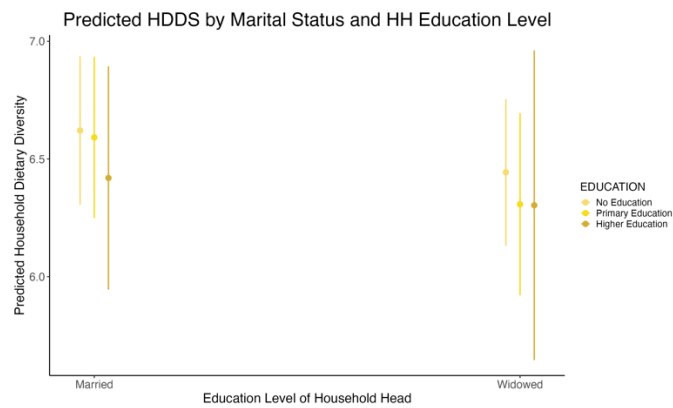
Figure 13. Predicted FCS by Marital Status and Education Level



For HDDS (Table 8), results are similarly null. Across all models, neither primary nor higher education significantly predicts HDDS. For instance, in Model 3.1, the coefficients for primary ($\beta = -0.12$, $p = 0.5$) and higher education ($\beta = -0.24$, $p = 0.3$) are small and statistically insignificant. In Model 3.5, effects are near zero ($\beta = -0.10$ to 0.06) and interaction terms remain non-significant ($\beta = -0.11$, $p = 0.7$; $\beta = 0.06$, $p = 0.9$), suggesting that any disadvantage in dietary diversity for widowed households is modest.

Figure 13 visually supports this conclusion. Predicted HDDS scores are similar across education levels for both marital groups, with overlapping confidence intervals and minimal variation. As with the previous models, widowed households tend to have slightly lower dietary diversity, but the differences are small and not systematically shaped by education level. The flat trend lines and broad error bars likely reflect both the limited sample of educated widowed women and the broader structural constraints in conflict-affected contexts, where gains from education may not translate into tangible improvements in dietary quality.

Figure 14. Predicted HDDS by Marital Status and Education Level



Together, these results indicate that education does not serve as a significant protective factor against food insecurity in this setting, nor does it provide additional benefits for widowed women. While the direction of some coefficients aligns with theoretical expectations, the lack of statistical significance and consistency across models points to broader structural barriers. Widespread educational disruption, limited employment options and entrenched gender norms likely prevent women from translating education into improved household outcomes.

Finally, the sample itself imposes limitations. Few women report formal education and even fewer widowed heads have higher education, reducing the statistical power to detect meaningful interaction effects. As such, although education is widely seen as a pathway to empowerment, its impact here appears conditional on supportive institutional and economic structures. *H3a* receives minimal support, while *H3b* is not supported by the data.

9. Discussion and Conclusion

9.1 Limitations

While this study offers valuable insights into the relationship between women's empowerment, marital status and household food security in Syria, several limitations should be acknowledged.

Internal Validity: While pooled OLS regression is suitable for exploratory analysis, it does not fully account for the repeated-measures structure of the panel data. Although clustered standard errors were used to partially address intra-household correlation, the model does not control for unobserved, time-invariant household characteristics, increasing the risk of omitted variable bias. This is particularly relevant in a context where stable household traits, likely shape food security outcomes over time. Data constraints limited the feasibility of fixed or random effects models, necessitating the use of pooled OLS. Multicollinearity checks indicated no significant concern; however, key predictors such as household income and size were excluded due to collinearity. The absence of women's personal income data further weakens the models' explanatory power.

Model performance, as measured by R^2 , was modest: 28% for the RCSI model, but only 9.5% and 3.5% for the FCS and HDDS models, respectively. These low values suggest limited explanatory power and highlight the need for more comprehensive data or improved model specification. Finally, the assumption of homogeneity across households may oversimplify underlying variation; in this regard, fixed or random effects models might have better captured unobserved heterogeneity if data allowed.

Measurement Limitations: While marital status offers a valuable proxy for empowerment in Syria's patriarchal context, it is an imperfect measure. Not all widows are empowered, and not all married women are disempowered. The exclusion of divorced and single women, though methodologically necessary, limits the generalizability of findings. Additionally, the context-specific meaning of marriage and widowhood may not be directly comparable to other settings, constraining the broader applicability of the empowerment framework.

The food security indicators used also have important limitations. Neither HDDS nor RCSI have been formally validated and all three indicators face limitations in capturing the cultural and context-specific dimensions of food security in Syria.

Sample Size and External Validity: The final analytical sample comprises 932 FHHs, a relatively small subset of the total population, limiting the statistical power of subgroup analyses. As the study focuses exclusively on FHHs, findings may not capture the dynamics of other type of households. Additionally, the data were collected only from government-controlled areas, which have been relatively less affected by conflict compared to opposition-held or heavily contested zones. As such, the findings may not be generalizable to all of Syria, especially areas where institutional collapse and displacement are more severe.

9.2 Policy Implications and Recommendations

The findings have several important policy implications for development actors and humanitarian organizations working in conflict-affected contexts:

1. Targeted Land Reform and Access Programs: Land ownership significantly improves food security, particularly for widowed women. Programs that facilitate secure land access or ownership for FHHs should be prioritized in post-conflict recovery planning, alongside legal reforms that guarantee women's rights to own, inherit and utilize land independently of male relatives.
2. Beyond Headship: Female headship does not equal empowerment. Ensuring widows' recognition as economic agents requires legal, institutional and economic reforms, as well as social protection that includes legal aid, resource access and community-level rights recognition.
3. Education-to-Empowerment Linkages: Given the limited direct impact of education on food security in this context, education programs must be linked with labor market access and structural reforms. Investments in women's education should be complemented by policies that improve employment opportunities and reduce gender-based discrimination in public life.

4. Localized and Context-Sensitive Interventions: Food security programming must be sensitive to regional and marital status-related disparities. Widowed women in areas such as Hama and Homs face particularly high vulnerabilities and should be prioritized in aid targeting and empowerment programming.

9.3 Conclusion

This thesis demonstrates that in patriarchal, conflict-affected contexts such as Syria, women's empowerment should be conceptualized not as a static condition, but as a relational and resource-contingent process in which marital status plays a pivotal role. Widowed women, though often holding formal authority, face structural disadvantages that undermine their empowerment and heighten their vulnerability. Yet, when provided with tangible assets such as land, they can convert this autonomy into meaningful resilience.

Due to data constraints, this analysis could not fully leverage the available panel data. Nonetheless, it marks an important step toward building empirical foundations for studying resilience and food security in fragile contexts. The literature has yet to adequately address how life shocks, such as partner loss, can disrupt empowerment trajectories and limit women's capacity to provide for their households. By highlighting this gap, the study not only offers immediate insights but also calls for further research into the interplay between gender, household food security and resilience in settings marked by protracted conflict and institutional breakdown.

In conclusion, empowering women is not merely about assigning them responsibility, but about enabling them to act with the resources, rights and recognition required to improve their families' welfare. As such, the pursuit of SDG2 must be seen in tandem with SDG5, particularly in fragile environments where empowerment pathways are complex, contested, and contextually shaped. For both policy and academic communities, this study offers a starting point for rethinking how gender, empowerment and food security intersect under conditions of structural constraint and societal transformation.

Appendix

Figure 15. FCS calculation example

Food groups	Frequency of consumption	Weight	Frequency × Weight
Cereals, grains, roots, and tubers	7	2	14
Legumes/pulses, nuts, and seeds	2	3	6
Milk and other dairy products	3	4	12
Meat, fish, and eggs	2	4	8
Vegetables and leaves	2	1	2
Fruits	1	1	1
Oil/fat/butter	7	0.5	3.5
Sugar or sweets	7	0.5	3.5
Condiments	7	0	0
Sum of score			50

Equation 1. Household Dietary Diversity Formula by the USAID 2006

$$\text{HDDS} = \sum_{i=1}^n I_i \quad (1)$$

Where:

- n is the total number of food groups (commonly 12).
- I_i is a binary indicator for food group i , where:

$$I_i = \begin{cases} 1 & \text{if food group } i \text{ was consumed in the reference period} \\ 0 & \text{otherwise} \end{cases}$$

Equation 2 Food Consumption Score Formula by the WFP

$$\text{FCS} = \sum_{i=1}^n f_i \times w_i \quad (2)$$

Where:

- f_i = frequency of consumption (in days) of food group i over the past 7 days
- w_i = weight assigned to food group i based on its nutritional value
- n = total number of food groups (typically 8)

Equation 3. Reduced Coping Strategy Index by the WFP 2019

$$\text{rCSI} = \sum_{i=1}^n f_i \times w_i \quad (3)$$

Where:

- f_i = frequency (in days) of coping strategy i practiced over the past 7 days
- w_i = severity weight assigned to coping strategy i
- n = total number of coping strategies (typically 5)

Table 3. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Land Ownership (RCSI Model)

Interaction Model Results for RCSI (Clustered SEs)															
Characteristic	Model 1.1			Model 1.2			Model 1.3			Model 1.4			Model 1.5		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
Marital Status of Household Head															
Married	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Widowed	0.86	-0.39, 2.1	0.2	0.74	-0.36, 1.8	0.2	0.73	-0.39, 1.9	0.2	0.82	-0.32, 2.0	0.2	0.86	-0.28, 2.0	0.14
Log Household Land Size Owned	-1.2	-1.8, -0.67	<0.001	-0.39	-0.89, 0.10	0.12	-0.40	-0.89, 0.10	0.12	-0.39	-0.88, 0.11	0.12	-0.38	-0.88, 0.11	0.13
Interaction: Marital Status × Log HH Land Size Owned															
Widowed * Log Household Land Size Owned	0.25	-0.57, 1.1	0.6	-0.09	-0.81, 0.62	0.8	-0.09	-0.81, 0.63	0.8	-0.09	-0.81, 0.62	0.8	-0.12	-0.83, 0.60	0.7
Governorate of Residence															
Aleppo	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Deir-ez-Zor	—	—	—	2.0	0.81, 3.2	0.001	2.0	0.79, 3.2	0.001	2.1	0.85, 3.3	<0.001	2.2	0.98, 3.4	<0.001
Hama	—	—	—	6.6	5.5, 7.7	<0.001	6.6	5.5, 7.8	<0.001	6.8	5.6, 8.0	<0.001	6.7	5.5, 7.9	<0.001
Homs	—	—	—	7.8	6.8, 8.8	<0.001	7.8	6.7, 8.9	<0.001	8.0	6.9, 9.1	<0.001	7.9	6.8, 9.0	<0.001
Rural Damascus	—	—	—	5.4	3.8, 6.9	<0.001	5.4	3.8, 6.9	<0.001	5.5	4.0, 7.1	<0.001	6.0	4.4, 7.6	<0.001
Education Level of Household Head															
No Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Primary Education	—	—	—	—	—	—	-0.06	-0.93, 0.81	0.9	-0.28	-1.2, 0.68	0.6	-0.21	-1.2, 0.75	0.7
Higher Education	—	—	—	—	—	—	0.08	-1.2, 1.3	0.9	-0.13	-1.4, 1.2	0.8	-0.02	-1.3, 1.3	>0.9
Age of Household Head	—	—	—	—	—	—	—	—	—	-0.02	-0.05, 0.02	0.3	-0.02	-0.05, 0.02	0.4
Receipt of Economic Aid	—	—	—	—	—	—	—	—	—	—	—	—	2.2	0.23, 4.2	0.029
Abbreviation: CI = Confidence Interval															

Table 4. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Land Ownership (FCS Model)

Interaction Model Results for FCS (Clustered SEs)															
Characteristic	Model 2.1			Model 2.2			Model 2.3			Model 2.4			Model 2.5		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
Marital Status of Household Head															
Married	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Widowed	-5.2	-8.5, -1.8	0.003	-3.9	-7.2, -0.64	0.019	-5.0	-8.4, -1.7	0.003	-5.6	-9.0, -2.2	0.001	-5.4	-8.8, -2.0	0.002
Log Household Land Size Owned	0.59	-0.87, 2.1	0.4	0.35	-1.1, 1.8	0.6	0.35	-1.1, 1.8	0.6	0.30	-1.2, 1.8	0.7	0.32	-1.1, 1.8	0.7
Interaction: Marital Status × Log HH Land Size Owned															
Widowed * Log Household Land Size Owned	1.8	-0.42, 3.9	0.11	1.8	-0.36, 3.9	0.10	1.9	-0.27, 4.0	0.087	1.9	-0.23, 4.0	0.081	1.8	-0.34, 3.9	0.10
Governorate of Residence															
Aleppo	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Deir-ez-Zor	—	—	—	-5.8	-9.4, -2.2	0.002	-5.1	-8.7, -1.5	0.005	-5.5	-9.1, -1.9	0.003	-4.8	-8.4, -1.2	0.009
Hama	—	—	—	-9.5	-13, -6.3	<0.001	-7.7	-11, -4.3	<0.001	-8.7	-12, -5.2	<0.001	-9.2	-13, -5.7	<0.001
Homs	—	—	—	-0.45	-3.4, 2.5	0.8	1.6	-1.6, 4.9	0.3	0.72	-2.6, 4.1	0.7	0.23	-3.1, 3.6	0.9
Rural Damascus	—	—	—	-1.1	-5.7, 3.5	0.6	-0.53	-5.1, 4.1	0.8	-1.5	-6.2, 3.1	0.5	0.62	-4.2, 5.4	0.8
Education Level of Household Head															
No Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Primary Education	—	—	—	—	—	—	-3.2	-5.8, -0.58	0.016	-1.8	-4.6, 1.1	0.2	-1.4	-4.3, 1.4	0.3
Higher Education	—	—	—	—	—	—	-5.2	-8.9, -1.4	0.007	-3.8	-7.7, 0.06	0.054	-3.3	-7.2, 0.59	0.10
Age of Household Head	—	—	—	—	—	—	—	—	—	0.12	0.01, 0.22	0.025	0.13	0.03, 0.23	0.013
Receipt of Economic Aid	—	—	—	—	—	—	—	—	—	—	—	—	11	5.2, 17	<0.001
Abbreviation: CI = Confidence Interval															

Table 5. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Land Ownership (HDDS Model)

Interaction Model Results for HDDS (Clustered SEs)														
Characteristic	Model 3.1			Model 3.2			Model 3.3			Model 3.4			Model 3.5	
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI
Marital Status of Household Head														
Married	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Widowed	-0.53	-0.87, -0.18	0.003	-0.46	-0.81, -0.11	0.010	-0.52	-0.88, -0.17	0.004	-0.55	-0.91, -0.19	0.003	-0.55	-0.91, -0.19
Log Household Land Size Owned	0.01	-0.15, 0.16	>0.9	0.02	-0.14, 0.17	0.8	0.02	-0.14, 0.17	0.8	0.01	-0.14, 0.17	0.9	0.01	-0.14, 0.17
Interaction: Marital Status × Log HH Land Size Owned														
Widowed * Log Household Land Size Owned	0.28	0.05, 0.51	0.016	0.28	0.05, 0.51	0.016	0.28	0.06, 0.51	0.014	0.29	0.06, 0.51	0.014	0.28	0.06, 0.51
Governorate of Residence														
Aleppo	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Deir-ez-Zor	—	—	—	0.18	-0.20, 0.56	0.4	0.22	-0.17, 0.60	0.3	0.20	-0.19, 0.58	0.3	0.22	-0.17, 0.61
Hama	—	—	—	-0.41	-0.75, -0.07	0.018	-0.31	-0.67, 0.06	0.10	-0.36	-0.74, 0.01	0.058	-0.38	-0.76, 0.00
Homs	—	—	—	0.11	-0.20, 0.43	0.5	0.23	-0.11, 0.58	0.2	0.18	-0.17, 0.54	0.3	0.17	-0.19, 0.52
Rural Damascus	—	—	—	-0.08	-0.56, 0.41	0.8	-0.05	-0.53, 0.44	0.9	-0.10	-0.60, 0.39	0.7	-0.03	-0.55, 0.48
Education Level of Household Head														
No Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Primary Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Higher Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Age of Household Head	—	—	—	—	—	—	—	—	—	0.01	0.00, 0.02	0.2	0.01	0.00, 0.02
Receipt of Economic Aid	—	—	—	—	—	—	—	—	—	—	—	—	0.35	-0.28, 0.97
Abbreviation: CI = Confidence Interval														

Table 4. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Education Level (RCSI Model)

Interaction Model Results for RCSI (Clustered SEs)															
Characteristic	Model 1.1			Model 1.2			Model 1.3			Model 1.4			Model 1.5		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
Marital Status of Household Head															
Married	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Widowed	2.6	1.3, 3.9	<0.001	1.1	-0.12, 2.3	0.078	1.2	0.00, 2.4	0.049	1.1	-0.09, 2.3	0.069	1.2	-0.06, 2.4	0.063
Education Level of Household Head															
No Education	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Primary Education	3.3	2.0, 4.6	<0.001	0.32	-0.89, 1.5	0.6	0.11	-1.1, 1.4	0.9	0.09	-1.2, 1.3	0.9	0.18	-1.1, 1.4	0.8
Higher Education	4.4	2.8, 6.1	<0.001	0.25	-1.3, 1.8	0.8	0.06	-1.5, 1.7	>0.9	0.11	-1.5, 1.7	0.9	0.22	-1.4, 1.8	0.8
Interaction: Marital Status × HH Education Level															
Widowed * Primary Education	-0.86	-2.6, 0.93	0.3	-0.76	-2.4, 0.84	0.4	-0.84	-2.4, 0.76	0.3	-0.74	-2.3, 0.86	0.4	-0.78	-2.4, 0.82	0.3
Widowed * Higher Education	-1.4	-4.1, 1.4	0.3	-0.31	-2.8, 2.2	0.8	-0.45	-2.9, 2.0	0.7	-0.45	-2.9, 2.0	0.7	-0.43	-2.9, 2.0	0.7
Governorate of Residence															
Aleppo	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Deir-ez-Zor	—	—	—	2.2	1.0, 3.4	<0.001	2.3	1.1, 3.5	<0.001	2.0	0.77, 3.2	0.001	2.1	0.91, 3.4	<0.001
Hama	—	—	—	6.8	5.7, 8.0	<0.001	7.0	5.8, 8.2	<0.001	6.8	5.6, 8.0	<0.001	6.7	5.5, 7.9	<0.001
Homs	—	—	—	8.1	7.0, 9.2	<0.001	8.3	7.2, 9.4	<0.001	8.0	6.8, 9.1	<0.001	7.9	6.7, 9.0	<0.001
Rural Damascus	—	—	—	5.6	4.1, 7.2	<0.001	5.8	4.3, 7.4	<0.001	5.5	3.9, 7.1	<0.001	5.9	4.3, 7.5	<0.001
Age of Household Head															
Age of Household Head	—	—	—	—	—	—	-0.02	-0.06, 0.01	0.2	-0.02	-0.05, 0.01	0.3	-0.02	-0.05, 0.02	0.3
Log Household Land Size Owned															
Log Household Land Size Owned	—	—	—	—	—	—	—	—	—	-0.42	-0.79, -0.05	0.026	-0.43	-0.80, -0.06	0.024
Receipt of Economic Aid															
Receipt of Economic Aid	—	—	—	—	—	—	—	—	—	—	—	—	2.2	0.24, 4.2	0.028
Abbreviation: CI = Confidence Interval															

Table 5. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Education Level (FCS Model)

Interaction Model Results for FCS (Clustered SEs)															
Characteristic	Model 2.1			Model 2.2			Model 2.3			Model 2.4			Model 2.5		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
Marital Status of Household Head															
Married	—	—		—	—		—	—		—	—		—	—	
Widowed	-2.8	-6.3, 0.72	0.12	-1.1	-4.6, 2.5	0.6	-1.9	-5.5, 1.8	0.3	-1.6	-5.2, 2.0	0.4	-1.5	-5.1, 2.1	0.4
Education Level of Household Head															
No Education	—	—		—	—		—	—		—	—		—	—	
Primary Education	-1.8	-5.3, 1.7	0.3	-1.1	-4.7, 2.5	0.5	-0.04	-3.8, 3.7	>0.9	0.02	-3.7, 3.7	>0.9	0.47	-3.2, 4.2	0.8
Higher Education	-4.9	-9.3, -0.43	0.032	-4.0	-8.6, 0.69	0.095	-3.0	-7.7, 1.8	0.2	-3.1	-7.8, 1.6	0.2	-2.6	-7.3, 2.1	0.3
Interaction: Marital Status × HH Education Level															
Widowed * Primary Education	-2.8	-7.6, 2.0	0.3	-3.6	-8.4, 1.1	0.13	-3.2	-7.9, 1.6	0.2	-3.5	-8.2, 1.3	0.2	-3.7	-8.4, 1.1	0.13
Widowed * Higher Education	-1.1	-8.6, 6.4	0.8	-1.5	-8.8, 5.8	0.7	-0.77	-8.1, 6.6	0.8	-0.77	-8.1, 6.6	0.8	-0.66	-7.9, 6.6	0.9
Governorate of Residence															
Aleppo	—	—		—	—		—	—		—	—		—	—	
Deir-ez-Zor	—	—		-6.2	-9.8, -2.6	<0.001	-6.5	-10, -2.9	<0.001	-5.7	-9.4, -2.0	0.002	-5.0	-8.6, -1.4	0.007
Hama	—	—		-8.3	-12, -4.9	<0.001	-9.3	-13, -5.8	<0.001	-8.6	-12, -5.1	<0.001	-9.1	-13, -5.6	<0.001
Homs	—	—		0.84	-2.3, 4.0	0.6	0.00	-3.3, 3.3	>0.9	0.86	-2.5, 4.2	0.6	0.36	-3.0, 3.7	0.8
Rural Damascus	—	—		-1.6	-6.1, 3.0	0.5	-2.5	-7.1, 2.1	0.3	-1.6	-6.3, 3.1	0.5	0.58	-4.2, 5.4	0.8
Age of Household Head	—	—		—	—		0.11	0.01, 0.21	0.030	0.11	0.01, 0.21	0.036	0.12	0.02, 0.22	0.020
Log Household Land Size Owned	—	—		—	—		—	—		1.2	0.11, 2.3	0.031	1.2	0.09, 2.3	0.034
Receipt of Economic Aid	—	—		—	—		—	—		—	—		11	5.4, 17	<0.001
Abbreviation: CI = Confidence Interval															

Table 6. Pooled OLS Regression Summary Table with Interaction Terms for Marital Status and Education Level (HDDS Model)

Interaction Model Results for HDDS (Clustered SEs)															
Characteristic	Model 3.1			Model 3.2			Model 3.3			Model 3.4			Model 3.5		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
Marital Status of Household Head															
Married	—	—		—	—		—	—		—	—		—	—	
Widowed	-0.21	-0.58, 0.16	0.3	-0.17	-0.54, 0.21	0.4	-0.21	-0.60, 0.17	0.3	-0.18	-0.57, 0.20	0.4	-0.18	-0.56, 0.21	0.4
Education Level of Household Head															
No Education	—	—		—	—		—	—		—	—		—	—	
Primary Education	-0.12	-0.49, 0.24	0.5	-0.12	-0.50, 0.27	0.6	-0.05	-0.45, 0.35	0.8	-0.04	-0.44, 0.35	0.8	-0.03	-0.43, 0.37	0.9
Higher Education	-0.24	-0.71, 0.22	0.3	-0.26	-0.76, 0.23	0.3	-0.20	-0.71, 0.30	0.4	-0.22	-0.72, 0.28	0.4	-0.20	-0.71, 0.30	0.4
Interaction: Marital Status × HH Education Level															
Widowed * Primary Education	-0.12	-0.62, 0.39	0.7	-0.09	-0.60, 0.41	0.7	-0.07	-0.57, 0.44	0.8	-0.10	-0.61, 0.41	0.7	-0.11	-0.61, 0.40	0.7
Widowed * Higher Education	-0.05	-0.83, 0.73	>0.9	0.01	-0.77, 0.80	>0.9	0.06	-0.73, 0.84	0.9	0.06	-0.72, 0.84	0.9	0.06	-0.72, 0.84	0.9
Governorate of Residence															
Aleppo				—	—		—	—		—	—		—	—	
Deir-ez-Zor				0.12	-0.26, 0.50	0.5	0.11	-0.28, 0.49	0.6	0.20	-0.19, 0.59	0.3	0.22	-0.17, 0.61	0.3
Hama				-0.37	-0.74, -0.01	0.042	-0.43	-0.80, -0.06	0.024	-0.35	-0.73, 0.02	0.067	-0.37	-0.75, 0.01	0.055
Homs				0.15	-0.19, 0.49	0.4	0.10	-0.25, 0.45	0.6	0.20	-0.15, 0.56	0.3	0.19	-0.17, 0.54	0.3
Rural Damascus				-0.15	-0.63, 0.34	0.6	-0.20	-0.69, 0.29	0.4	-0.09	-0.59, 0.41	0.7	-0.02	-0.53, 0.50	>0.9
Age of Household Head							0.01	0.00, 0.02	0.2	0.01	0.00, 0.02	0.3	0.01	0.00, 0.02	0.2
Log Household Land Size Owned										0.15	0.03, 0.26	0.016	0.14	0.03, 0.26	0.016
Receipt of Economic Aid													0.38	-0.25, 1.0	0.2
Abbreviation: CI = Confidence Interval															

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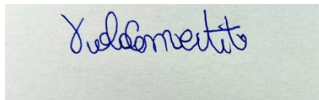
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I acknowledge the use of artificial intelligence-based tools, especially OpenAI's ChatGPT-4, in the development of this thesis. These tools were employed to support tasks such as identifying synonyms, rephrasing sentences and paragraphs, and assisting with LaTeX formatting and R coding. All AI-generated content was meticulously reviewed, edited and verified by me to ensure accuracy, clarity and appropriateness.

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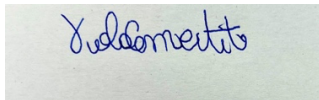
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